A Joint Optimization Method for High-Resolution Wide-Swath SAR Imaging: Combining Signal Transmitting and Imaging Perspectives

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***Abstract*—Synthetic aperture radars (SARs) require a larger pulse repetition interval (PRI) to achieve a greater range swath width, which is often addressed by designing nonuniform pulse transmission patterns. However, decoupling the pulse transmission pattern and SAR imaging algorithms in existing works hinders the full utilization of the antialiasing capability of nonuniform transmission during the imaging procedure. To address this issue, this article proposes a joint optimization method of learnable pulse transmission pattern and SAR imaging algorithm, namely, JointSAR. Specifically, we optimize the pulse transmission intervals in a random parameter freezing way and develop a deep unfolding network (DUN) based on an approximate observation model. Simulation and measured data-based experiments verify that the proposed method achieves better SAR imaging quality and maintains wide-swath imaging capability in comparison with the existing state-of-the-art methods.**

***Index Terms*—Deep unfolding network (DUN), high resolution, nonuniform sampling, synthetic aperture radar (SAR) imaging, wide swath.**

I. Introduction

SYNTHETIC aperture radar (SAR) imaging, with its inherent immunity to temporal and climatic influences, has obtained extensive attention for its capability to observe targets continuously throughout the day and under all-weather conditions [1], [2], [3]. In many practical applications, there is often a desire to achieve both a wide imaging swath and superior azimuthal resolution at the same time. However, in the realm of SAR, an inherent contradiction typically exists between the aforementioned two purposes. According to the Nyquist sampling theorem [4], enhancing the azimuthal resolution often leads to an increase in the pulse repetition frequency (PRF). On the other hand, the range swath width depends on the duration of the received echo, demanding an augmentation in the pulse repetition interval (PRI, a reciprocal of PRF) to obtain the larger range swath widths. Therefore, we often have to balance between azimuthal resolution and range swath width following the Nyquist sampling theorem [4].

A. Wide-Swath SAR Imaging

To achieve a wider SAR imaging swath, some research works use multichannel digital beamforming [5], [6], [7] or multiple-aperture configuration [8], but with significantly high complexity and cost. For the cheap single-channel SAR system, PRF needs to be decreased for a wider swath of SAR imaging. The simplest idea is to reduce PRF and perform uniform downsampling in the azimuth direction but with signal aliasing according to the Nyquist sampling theorem. To address this issue, recent research works have investigated the antialiasing capability of nonuniform pulse transmission patterns in the context of the SAR imaging [9], [10], [11].

Simple random selection pulse transmission, which randomly selects a subset from a uniform pulse sequence, is a commonly used form of nonuniform transmission pattern. Many studies utilize the linear random projection or partial identity matrix to realize simple random sampling [12], [13], [14], [15], which effectively suppresses aliasing and preserves original information while reducing data volume. However, since simple random sampling only changes the sampling rate without altering the minimum transmission interval, it cannot be used to expand the swath width. To achieve wide-swath SAR imaging, Yu et al. [16] proposed a time-space transmission method named AgileSAR, dividing the swath into multiple subswaths in the space domain to obtain a wider imaging area, while the elaborate design of multiple subswaths also makes the system complicated.

Some research works [17], [18] consider an innovative concept called staggered SAR, where PRI is continuously varied in a linear manner. This approach enables wide-swath imaging by increasing the minimum PRI and helps to mitigate aliasing effects.Due to the excellent performance of Jittered sampling and Poisson disk sampling in reducing aliasing [19], [20], several studies have explored the capability of nonuniform sampling in various radar systems [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], such as

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pulsed-Doppler radars [27], [28], [29], [30], [31], multi-in multi-out (MIMO) radars [33], [34], [35], [36], and wide-swath SARs [37], [38], [39].According to the analysis by Yang et al. [38], the antialiasing capability of Jittered transmission decreases as the swath width increases, and the Poisson disk transmission pattern can achieve better ambiguity suppression performance in comparison with the Jittered pattern.

The aforementioned staggered SAR, Jittered, and Poisson disk transmission patterns all fall under the category of continuously changing PRIs [17], [18], [37], [38], [39]. Although these approaches offer good antialiasing effects, it leads to the loss of received echo caused by the occlusion of the transmitted pulses, resulting in unavoidable blind area. Based on this limitation, Dong et al. [40] proposed a pulse transmission pattern known as random PRF variation, which maintains uniform transmission between consecutive pulses and changes the PRF periodically. However, the antialiasing performance by random PRF variation [40] is relatively sensitive to the selection of PRFs.

Although researchers working on designing nonuniform pulse transmission patterns also propose related SAR imaging methods [18], [38], [40], [41], they are not deep coupled and SAR imaging algorithms are not improved for specific nonuniform pulse transmission patterns, resulting in inadequate leverage of the antialiasing characteristics of nonuniform pulses and affecting the final SAR imaging quality.

B. Deep Unfolding Network (DUN)-Based SAR Imaging

In recent years, DUNs has been widely applied in the field of SAR imaging to enhance the imaging resolution and quality by exploiting the sparsity of the observed scenes even at significantly reduced sampling rates [12], [13], [14], [42], [43], [44], [45], [46], [47]. In general, DUN-based SAR imaging methods are often based on iterative algorithms in compressive sensing [48], such as the iterative shrinkage-thresholding algorithm (ISTA) [49], the approximate message passing (AMP) algorithm [50], and the alternating direction method of multipliers (ADMMs) algorithm [51]. By unfolding the layers of iterative algorithms and optimizing the threshold and sparsity as learnable parameters, DUN-based methods can achieve excellent iterative performance without manual parameter tuning. For example, Zhang and Ghanem [12] unrolled the ISTA algorithm into ISTA-Net and incorporated the complex-valued characteristics of SAR signals into complex matrix operations, providing high-quality SAR imaging results.

In addition, the size of the observation matrix in the compressive sensing is really huge in the SAR signal model, leading to huge matrix multiplications in iterative-based algorithms and resulting in high computational complexity. To tackle this issue, researchers have introduced an approximate observation method into iterations of SAR imaging [14], [38], [40], [52], [53], [54], [55]. Approximate observation method utilizes the reversibility of traditional matched filter-based SAR imaging algorithms in iterative-based algorithms, such as range–Doppler algorithm (RDA) and chirp scaling algorithm (CSA), to derive imaging and inverse imaging operators and replace the original large matrix multiplications, and significantly reduce the imaging algorithm complexity. Yang et al. [38] combined the nonuniform CSA imaging operator with the ISTA algorithm, and Dong et al. [40] integrated the Omega-K imaging operator with a fast ISTA algorithm to propose compressive sensing-based SAR imaging methods for nonuniform under-sampled echo data. Similarly, Kang et al. [14] integrated the RDA operator with a DUN based on the ISTA algorithm, reducing the computational cost of SAR imaging.

Moreover, compressive sensing-based DUN SAR imaging method typically requires the reconstructed scene to be sparse, but SAR images are not always sparse, which may make the reconstruction algorithm less effective. Some research efforts [13], [46], [47] aim to address this issue based on nonsparse image representation. NSR-NET [47] achieves the nonsparse SAR imaging by unrolling the proximal gradient descent algorithm into a DUN that employs 2-D convolutional layers and a learnable piecewise linear layer for image representation and regularization. An et al. [46] tackled the nonsparse SAR imaging problem by modeling it as a joint low-rank and sparse matrix recovery problem, then unrolling ADMMs into a DUN. However, these algorithms are often cumbersome and complex due to their reliance on intricate matrix operations, which can lead to longer processing times and ultimately reduce the overall efficiency of SAR imaging.

Existing DUN-based SAR imaging algorithms [12], [13], [14], [47] rely on simple random sampling to obtain under-sampled echoes and focus on high-resolution imaging, neglecting the design of nonuniform pulse transmission patterns and failing to achieve wide-swath imaging.

C. Contribution of Our Article

Existing SAR imaging algorithms often separately improve the pulse transmission patterns and the SAR imaging methods to obtain a wider swath or a higher imaging resolution [13], [14], [39] or simply cascade the both one to achieve the high-resolution wide-swath performance [18], [38], [40], [41]. Decoupling the pulse transmission pattern and SAR imaging methods will lead to inadequate utilization of the antialiasing and denoising characteristics of nonuniform pulse transmission patterns, thereby affecting the final SAR imaging quality.

In this article, we propose a joint optimization method, named JointSAR, for high-resolution and wide-swath SAR imaging, which couples the DUN-based SAR imaging algorithm with a learnable pulse transmission pattern to address the aforementioned issue. The proposed method allows the characteristics of nonuniform pulse transmission patterns to be considered in the design of SAR imaging algorithms, also makes the design of nonuniform pulse transmission patterns more suitable for the task of SAR imaging.

Specifically, in the learnable pulse transmission part, we learn the unfixed part of PRI and guarantee the minimum interval extension between adjacent pulses to achieve a wide swath. In the SAR imaging part, we propose a learnable DUN based on a convolutional neural network (CNN) mapping and CSA approximate observation operators to suppress aliasing and sidelobes. In addition, in order to improve the antialiasing capability of the learnable nonuniform pulse transmission timing, we introduce a random parameter freezing strategy in the optimization process to increase the robustness of the learned pulse transmission timing.

After learning the optimal pulse transmission patterns and SAR imaging algorithms through data-driven methodologies,

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SAR can transmit pulses according to these well-designed nonuniform sequences, and then, we use the learned SAR imaging algorithms to achieve high resolution and better SAR image quality under the same extended wide-swath conditions. Experimental results based on the simulated and measured SAR data demonstrate the superiority of our proposed JointSAR in comparison with existing state-of-the-art methods.

Note that the latest article [15] also explores the joint optimization of azimuth downsampling pattern and SAR imaging method.This work represents a representative study on nonuniform sampling design for SAR imaging. However, there is a significant difference between our approach and [15]. Note that we learn the transmission pattern to achieve wide-swath imaging, while [15] learns the SAR signal downsampling pattern to reduce data redundancy but keeps the initial minimum interval during the transmission process and is unable to achieve a wider swath. In addition, their generated sampling patterns include nearly zero sampling intervals, which would prevent any echo from being received in a single-transmit–single-receive SAR system, making their method infeasible for wide-swath applications.These distinctions reflect not just significant improvements over this latest work [15] but a fundamentally different approach to tackling the challenges of wide-swath SAR imaging, emphasizing our unique contribution to the field.

The remainder of this article is structured as follows. In Section II, we present the basic SAR signal model of SAR imaging. Section III elaborates the proposed JointSAR method. Section IV validates the performance of the proposed method through experimental results using simulated and measured SAR data. Section V concludes this article with some remarks and hints at plausible future research.

II. Preliminaries

A. SAR Signal Model

SAR moves relative to the Earth's surface at a certain speed, emitting pulse signals along a specific direction and receiving the echoes reflected from the Earth's surface. Pulses transmitted by the radar are typically linear frequency modulated, i.e.,

$$p(\tau) = \text{rect}\left(\frac{\tau}{T_p}\right)\exp\left\{j2\pi f_0\tau + j\pi K_r\tau^2\right\} \quad \tau \in \left[-\frac{T_p}{2}, \frac{T_p}{2}\right) \tag{1}$$

where p is the transmitted pulse, τ represents the range time, T_p stands for the pulse duration, f_0 denotes the carrier frequency of the transmitted pulses, K_r represents the frequency modulation rate of the pulse in the range direction, and rect$(\cdot)$ denotes the rectangular envelope function. After demodulation, sampling, and discretization, the received echo can be expressed as follows:

$$s\left(\tau_{n_r}, \eta_{n_a}\right) = \sum_{m_r=1}^{M_r}\sum_{m_a=1}^{M_a}\sigma\left(x_{m_a}, y_{m_r}\right) \cdot a\left[\tau_{n_r}, \eta_{n_a}, x_{m_a}, y_{m_r}\right] + N\left(\tau_{n_r}, \eta_{n_a}\right) \tag{2}$$

where η represents the azimuthal time, $s(\tau_{n_r}, \eta_{n_a})$ represents the n_rth sample of the n_ath pulse of the received echo s, M_r and M_a represent the number of discrete points in the range and azimuth directions, respectively, $\sigma(x_{m_a}, y_{m_r})$ denotes the scattering coefficient of the target point located at the m_ath coordinate in the azimuth direction, m_rth denotes the coordinate in the range direction, and $N(\tau_{n_r}, \eta_{n_a})$ represents the additive noise of the SAR receiving system. In (2), the detailed expression of $a[\tau_{n_r}, \eta_{n_a}, x_{m_a}, y_{m_r}]$ is

$$\begin{aligned} a\left[\tau_{n_r}, \eta_{n_a}, x_{m_a}, y_{m_r}\right] &= w_r\left(\tau_{n_r} - \frac{2R\left(x_{m_a}, y_{m_r}, \eta_{n_a}\right)}{c}\right) \\ &\cdot w_a\left(\eta_{n_a} - \frac{x_{m_a}}{V}\right) \\ &\cdot \exp\left\{-j\frac{4\pi f_0 R\left(x_{m_a}, y_{m_r}, \eta_{n_a}\right)}{c}\right\} \\ &\cdot \exp\left\{j\pi K_r\left(\tau_{n_r} - \frac{2R\left(x_{m_a}, y_{m_r}, \eta_{n_a}\right)}{c}\right)^2\right\} \end{aligned} \tag{3}$$

where w_r is the range envelope function, w_a is the azimuth envelope function, $R(x_{m_a}, y_{m_r}, \eta_{n_a})$ denotes the distance between the SAR sensor and the observed target point at (x_{m_a}, y_{m_r}) with the slow time η_{n_a}, and V is the equivalent radar velocity.

All the scattering coefficients $\{\sigma(x_{m_a}, y_{m_r}), m_a = 1, \ldots, M_a, m_r = 1, \ldots, M_r\}$ can be represented as $\mathbf{X} \in \mathbb{C}^{M_r \times M_a}$ and the received echo $\{s(\tau_{n_r}, \eta_{n_a}), n_r = 1, \ldots, N_r, n_a = 1, \ldots, \tilde{N}_a\}$ can be represented as $\mathbf{Y} \in \mathbb{C}^{\tilde{N}_a \times N_r}$, with $\tilde{N}_a$ and N_r being the number of pulses transmitted in the azimuth direction and the number of sampling points for each pulse in the range direction, respectively. In addition, the nonuniform pulse transmission timings are defined as $\mathbf{T}_\eta = [\eta_1, \ldots, \eta_{n_a}, \ldots, \eta_{\tilde{N}_a}] \in \mathbb{R}^{\tilde{N}_a}$. Then, (2) is expressed as follows:

$$\mathbf{Y} = \text{mat}\left(\mathbf{A}\left(\mathbf{T}_\eta, \mathbf{\Theta}\right)\text{vec}\left(\mathbf{X}\right)\right) + \mathbf{N} \tag{4}$$

where $\text{vec}(\mathbf{X}) \in \mathbb{C}^{M_r M_a \times 1}$ represents the vectorization results of matrices $\mathbf{X}$ and mat$(\cdot)$ is the opposite operation of vec$(\cdot)$. $A(\mathbf{T}_\eta, \mathbf{\Theta})$ is the observation matrix that changes with $\mathbf{T}_\eta$ and $\mathbf{\Theta}$, $\mathbf{\Theta}$ is the SAR system parameters, including the radar center frequency, bandwidth, and PRF. $\mathbf{N}$ is the Gaussian white noise matrix.

B. Classical SAR Imaging Algorithm

According to the compressive sensing theory [48] and approximate observation model [38], (4) can be solved by

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}}\left\{\frac{1}{2}\|\mathbf{Y} - G\left(\mathbf{X}\right)\|_2^2 + \kappa\|\mathbf{X}\|_1\right\} \tag{5}$$

where $G(\cdot)$ is an approximate observation model from $\mathbf{X}$ to $\mathbf{Y}$, $\|\cdot\|_2$ and $\|\cdot\|_1$ represents $L2$-norm and $L1$-norm, respectively, and κ is the regularization parameter. The solution of (5) can be obtained by the classic and well-known ISTA with approximate mapping model [38], [49], and the kth iteration is defined as follows:

$$\mathbf{X}^k = E_\mu\left(\mathbf{X}^{k-1} + \alpha H\left(\mathbf{Y} - G\left(\mathbf{X}^{k-1}\right)\right)\right) \tag{6}$$

where $H(\cdot)$ is the approximate mapping algorithm from $\mathbf{Y}$ to $\mathbf{X}$ and E_μ is the soft thresholding function [49]

$$E_\mu\left(\mathbf{X}\right) = \text{sgn}\left(\mathbf{X}\right)\cdot\max\left(|\mathbf{X}| - \mu, 0\right) \tag{7}$$

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where sgn(·) is the sign function and μ is the soft thresholding parameter. For $H(\cdot)$ and $G(\cdot)$ in (6), we can use traditional imaging operators based on matched filtering. Here, we introduce the classic CSA operator and its inverse operator [38], which also serves as the foundation for the subsequently proposed DUN imaging algorithm.

Overall, CSA primarily involves four fast Fourier transforms (FFTs) and three phase compensation processes, including chirp scaling operations, range compression, and azimuth compression. Since the traditional azimuth FFT cannot be performed directly in nonuniform azimuth echoes, a downsampled Fourier transform matrix is used in CSA. The CSA imaging operator is as follows:

$$\tilde{\mathbf{X}} = H(\mathbf{Y}) = \mathrm{IFFT}_\eta\left(\mathrm{IFFT}_\tau\left(\mathrm{FFT}_\tau\left(\left(\mathbf{F}_\eta\mathbf{Y}\right)\odot\mathbf{S}_c\right)\odot\mathbf{P}_\tau\right)\odot\mathbf{P}_\eta\right) \tag{8}$$

where $H(\cdot)$ denotes the CSA imaging operator and $\tilde{\mathbf{X}} \in \mathbb{C}^{M_a\times M_r}$ represents the image results. FFT_τ stands for the range FFT, $\mathbf{F}_\eta \in \mathbb{C}^{N_a\times\tilde{N}_a}$ represents the azimuth nonuniform Fourier transform matrix, which can be obtained by downsampling the Fourier transform matrix. IFFT_τ and IFFT_η denote the inverse range and azimuth FFTs, respectively. The matrices $\mathbf{S}_c \in \mathbb{C}^{N_a\times N_r}$, $\mathbf{P}_\tau \in \mathbb{C}^{N_a\times N_r}$, and $\mathbf{P}_\eta \in \mathbb{C}^{N_a\times N_r}$ represent the chirp scaling operation matrix, the range compression phase compensation matrix, and the azimuth compression phase compensation matrix, respectively. $\mathbf{S}_c$, $\mathbf{P}_\tau$, $\mathbf{P}_\eta$, and $\mathbf{F}_\eta$ are calculated based on the radar transmission timings $\mathbf{T}_\eta$ and radar parameters $\boldsymbol{\Theta}$, as detailed in [38] for a more detailed description.

Since each step in the CSA imaging operator except nonuniform FFT matrix multiplication is reversible, applying the inverse CSA operations to the imaging results can yield an approximate reconstruction of the original echo data, as shown in the following equation:

$$\tilde{\mathbf{Y}} = G(\mathbf{X}) = \mathbf{I}_\eta\left(\mathrm{IFFT}_\tau\left(\mathrm{FFT}_\tau\left(\mathrm{FFT}_\eta(\mathbf{X})\odot\overline{\mathbf{P}_\eta}\right)\odot\overline{\mathbf{P}_\tau}\right)\odot\overline{\mathbf{S}_c}\right) \tag{9}$$

where $\overline{\mathbf{S}_c}$, $\overline{\mathbf{P}_\tau}$, and $\overline{\mathbf{P}_\eta}$ are the complex conjugate matrices of $\mathbf{S}_c$, $\mathbf{P}_\tau$, and $\mathbf{P}_\eta$, respectively, and $\mathbf{I}_\eta$ is the nonuniform azimuth inverse Fourier transform matrix. Therefore, using the imaging operator $H(\cdot)$ and the echo simulation operator $G(\cdot)$ in (6), we can iteratively solve the SAR imaging problem in (5) and obtain the high-resolution SAR image.

C. Relationship Between Wide-Swath and Pulse Transmission Patterns

In single-channel SAR, there exists an inherent set of constraints between high azimuth resolution and wide swath width. According to the Nyquist sampling theorem, to avoid azimuth ambiguity, the azimuth sampling frequency must be greater than the Doppler bandwidth

$$\mathrm{PRF} \geq \alpha_o B_a \tag{10}$$

where α_o is the oversampling ratio in the azimuth direction and B_a is the Doppler bandwidth. Therefore, to achieve high azimuth resolution, it is necessary to increase the Doppler bandwidth, which means increasing the azimuth PRF.

On the other hand, to avoid range ambiguity, the range swath width W must satisfy

$$W \leq \left(\mathrm{PRI}_{\min} - 2T_p\right)\cdot\frac{c}{2}. \tag{11}$$

Thus, it is required to increase the minimum PRI to achieve a wide swath. The greater the extension of the minimum PRI, the greater the achievable swath width.

To achieve wide-swath imaging, existing methods [38], [40] increase $\mathrm{PRI}_{\min}$ to obtain a larger maximum unambiguous swath width and adopt nonuniform pulse transmission patterns to mitigate azimuth aliasing caused by downsampling. However, decoupling the pulse transmission pattern from the SAR imaging process fails to fully exploit the antialiasing and denoising benefits of nonuniform transmission, ultimately limiting the final imaging quality. Therefore, this work aims to find an optimal pulse transmission pattern to achieve swath width expansion while further improving the imaging quality.

III. Proposed JointSAR For High-Resolution and Wide-Swath SAR Imaging

In this section, we introduce the proposed JointSAR method with jointly optimized pulse transmission pattern and SAR imaging strategy. First, we introduce the learnable pulse transmission pattern; then, the DUN-based SAR imaging method, followed by the joint optimization method. Finally, the analysis of parameter quantity and computational complexity is elaborated.

Noted that the proposed JointSAR is first trained to generate an optimized pulse transmission pattern and the corresponding SAR imaging method. Fig. 1 shows the overall training flow architecture of the proposed JointSAR. First, observed scenarios of SAR are simulated randomly as ground truth (GT), and the nonuniform pulse transmission pattern is initialized randomly. In each iteration of parameter update, azimuthally nonuniform echoes are generated based on the current pulse transmission timings $\mathbf{T}_\eta$ and SAR system parameters $\boldsymbol{\Theta}$ according to the echo simulation model in (4). After that, the proposed DUN-based SAR imaging method is subsequently employed to produce SAR images, with learnable parameters such as the step size, threshold, and CNN parameters needing optimization. The SAR imaging results are then compared with the predefined GT to compute the reconstruction mean squared error (mse) loss, which together with the regularization terms for the jointly optimized pulse transmission pattern and imaging algorithm, forms the total loss function for the joint optimization. During backpropagation, a random parameter freezing strategy is proposed for the learnable pulse transmission intervals, which helps to prevent overfitting like dropout and enhances the randomness and robustness of this variable in the learning process. Finally, the Adam optimizer is utilized to iteratively learn the joint learnable variables, resulting in the optimal nonuniform pulse transmission timings and SAR imaging algorithm. After the training process is finished, the optimized pulse transmission pattern can be used in a practical SAR system to receive azimuthal nonuniform echo, and then, the SAR imaging result is obtained according to our learned SAR imaging method.

A. Learnable Pulse Transmission Pattern

This section introduces the proposed learnable pulse transmission pattern, as shown in Fig. 1.

First, we extracted paradigms including stratified, Poisson disk, Jittered, and other pulse transmission patterns in Table I,

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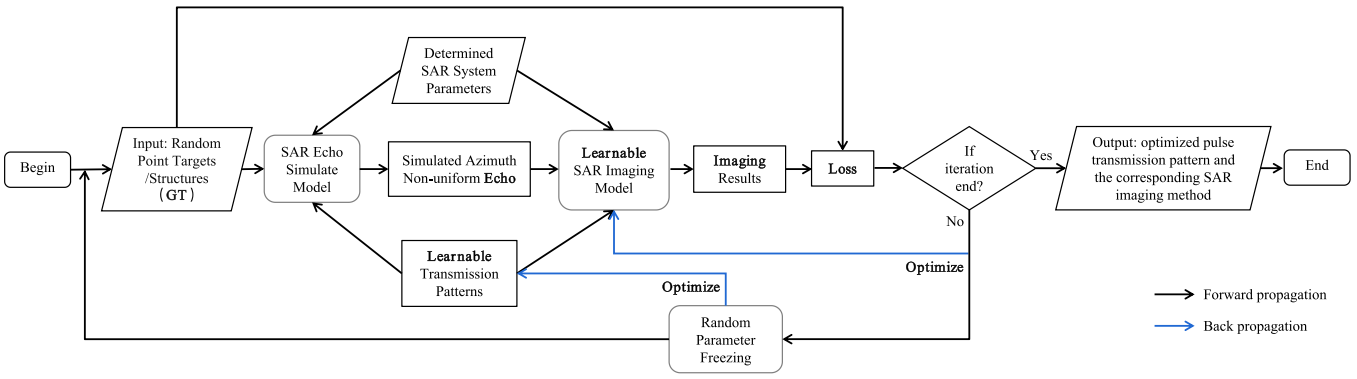

Fig. 1. Overall training flowchart of the proposed JointSAR.

TABLE I
FIXED AND RANDOM PARTS IN DIFFERENT PULSE TRANSMISSION MODES

Method	Fixed Part T_{fix}	Random Part d_{n_a}
Stratified Pattern	0	$Tri(0, T_{avg}, 2T_{avg})$
Jittered Pattern	$(1-\beta)T_{avg}$	$Tri(0, \beta T_{avg}, 2\beta T_{avg})$
Poisson disk Pattern	T_{min}	$Exp(\lambda)$

where T_{fix} and d_{n_a} represent the fixed and random part of pulse transmission intervals, respectively, T_{avg} and T_{min} represent the average and minimum pulse transmission intervals, respectively, and Tri(·) and Exp(·) mean the triangular distribution [56] and exponential distribution, respectively. Transmission intervals in all the existing transmission methods can be expressed as the sum of two parts: a fixed component T_{fix} and a random component d_{n_a}; then, the n_ath transmission interval $\eta_{\Delta_{n_a}}$ can be written as follows:

$$\eta_{\Delta_{n_a}} = T_{\text{fix}} + d_{n_a} \tag{12}$$

where the random part d_{n_a} has a minimum value of 0, thus $T_{\text{min}} = T_{\text{fix}}$ and wide-swath can be achieved by increasing T_{fix} to ensure the minimum pulse transmission interval. d_{n_a} ensures a degree of randomness in the pulse transmission timing, reducing aliasing in SAR images.

Based on (12), we propose a learnable pulse transmission pattern, where the random part of the pulse transmission intervals $\mathbf{d}_\eta = [d_1, \ldots, d_{n_a}, \ldots, d_{\tilde{N}_a}] \in \mathbb{R}^{\tilde{N}_a}$ in (12) are treated as learnable variables and T_{fix} is set as a multiple of the Nyquist minimum sampling interval, which corresponds to a multiple of the predetermined extended swath width. The n_ath element of the final transmission timing $\mathbf{T}_\eta$ can be expressed as follows:

$$\eta_{n_a} = n_a T_{\text{fix}} + \sum_{i=1}^{n_a} d_i. \tag{13}$$

In Section III-C, the loss function term to guide the learning of $\mathbf{d}_\eta$ will be introduced.

After obtaining the temporal pulse transmission timing $\mathbf{T}_\eta$ and $\mathbf{A}(\mathbf{T}_\eta, \boldsymbol{\Theta})$ can be calculated, and then, the received echo can be simulated based on (4) and predefined SAR observed scenario $\mathbf{x}$.

B. Deep Unfolded SAR Imaging Network Based on Approximate Observations

This section introduces the learnable SAR imaging model in Fig. 1. In general, we propose a deep unfolding SAR imaging strategy based on ISTA with approximate observations and CNN-based domain mapping.

The ISTA method with approximate observations has already been introduced in Section II-B. In order to reduce the number of iterations to speed up the imaging algorithm and more efficiently determine the step size, threshold, and other parameters in ISTA, we unfold the traditional iterative algorithm (6) into a DUN architecture, where each iteration becomes one layer in DUN. According to (6), one iteration in our SAR imaging algorithm can be divided into two steps

$$\begin{aligned} \mathbf{R}: \mathbf{Z}^k &= \mathbf{X}^{k-1} + \alpha^k \cdot H\left(\mathbf{Y} - G\left(\mathbf{X}^{k-1}\right)\right) \\ \mathbf{N}: \mathbf{X}^k &= E\left(\mathbf{Z}^k\right). \end{aligned} \tag{14}$$

As shown in (14), an intermediate imaging result $\mathbf{Z}^k$ is obtained through the linear approximate observation $\mathbf{R}$ module. Then, a sparse SAR imaging result is generated by a nonlinear $\mathbf{N}$ module.

For the linear $\mathbf{R}$ module in (14), the detailed structure of the kth iteration is visualized in Fig. 2. $\mathbf{X}^{k-1}$ is the imaging result from the previous iteration, and $\mathbf{Z}^k$ is the output of the $\mathbf{R}$ module. α^k is the only learnable parameter in the $\mathbf{R}$ module and is not shared across layers, enabling flexible adjustment of the step size at different iteration levels. $H(\cdot)$ and $G(\cdot)$, given by (8) and (9), respectively, represent the CSA imaging operator and its corresponding inverse operator. $\mathbf{S}_c$, $\mathbf{P}_\tau$, $\mathbf{P}_\eta$, $\mathbf{F}_\eta$, and $\mathbf{I}_\eta$ in Fig. 2 are calculated based on the radar transmission timings $\mathbf{T}_\eta$ and radar parameters $\boldsymbol{\Theta}$, see [38]. Using approximate observation operators $H(\cdot)$ and $G(\cdot)$, the linear $\mathbf{R}$ module corrects the results of the previous iteration based on the received echo $\mathbf{Y}$ to obtain a more refined imaging result. The inclusion of both forward and inverse operators ensures that the updates are guided by the physical principles of SAR imaging, making the method robust, efficient, and interpretable. Furthermore, this iterative correction ensures that each updated imaging result $\mathbf{Z}^k$ progressively aligns with the true scene.

For the nonlinear $\mathbf{N}$ module in (14), its main purpose is to remove noise-like artifacts caused by nonuniform pulse

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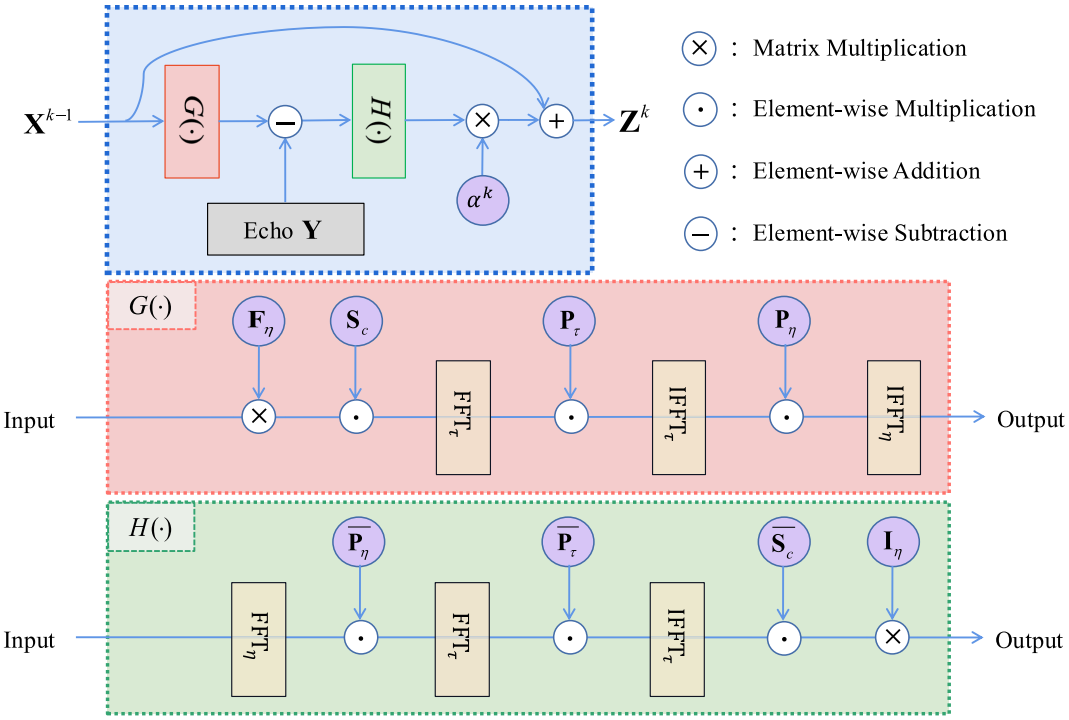

Fig. 2. Detailed framework diagram of the kth iteration of linear **R** module in (14).

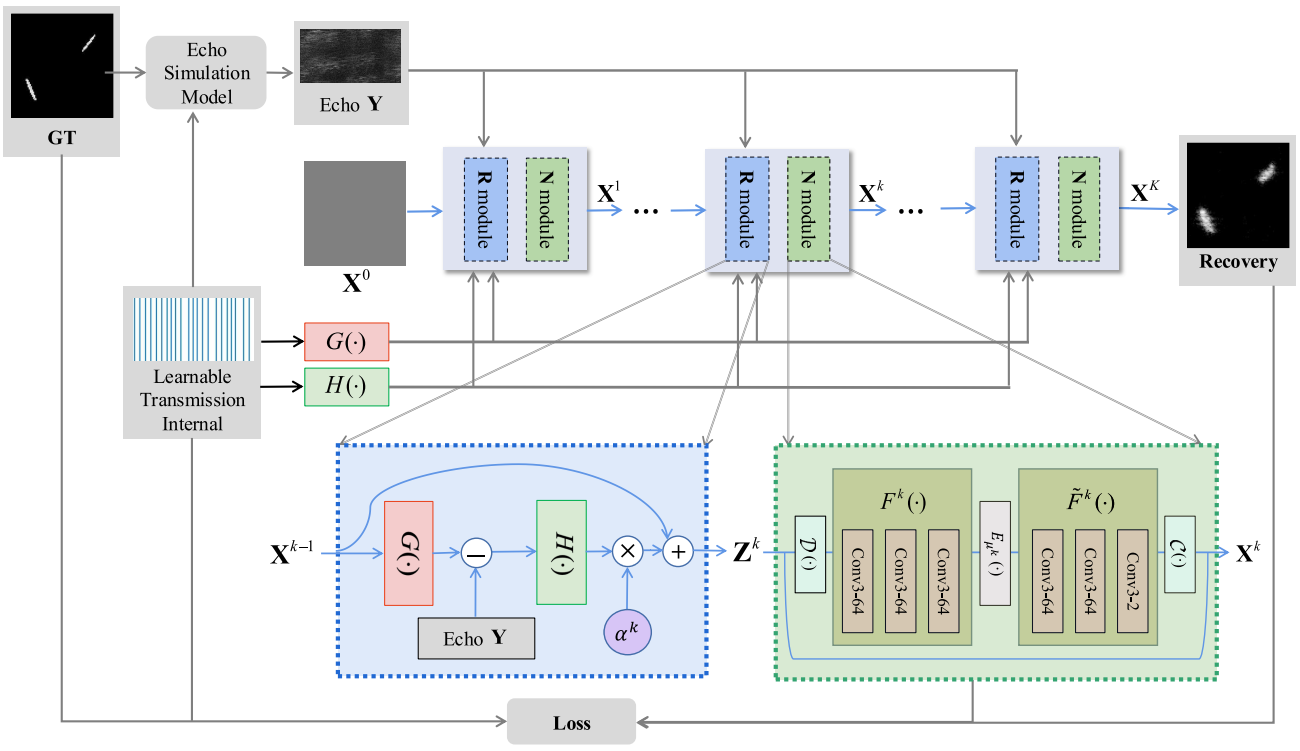

Fig. 3. Detailed framework diagram of our JointSAR with learnable pulse transmission pattern and SAR imaging method.

transmission pattern. We use CNNs to map the SAR images to a sparse domain, where the soft threshold operator is performed to scale data and reduce noise-like artifacts, and the SAR images are then mapped back to the original image domain. The specific formula of the **N** module is designed as follows:

$$\mathbf{X}^k = E\left(\mathbf{Z}^k\right) = \mathbf{Z}^k + \mathcal{C}\left(\tilde{F}^k\left(E_{\mu^k}\left(F^k\left(\mathcal{D}\left(\mathbf{Z}^k\right)\right)\right)\right)\right) \tag{15}$$

where E_{μ^k} is the soft thresholding function defined in (7) and the learnable parameter μ_k is trained without sharing

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parameters in each layer, $\mathcal{D}(\cdot)$ operator converts the complex matrix $\mathbf{Z}^k \in \mathbb{C}^{H\times W}$ into a real matrix with the size of $\mathbb{R}^{2\times H\times W}$. $\mathcal{C}(\cdot)$ operator in (15) performs the inverse transformation of $\mathcal{D}(\cdot)$

$$\begin{aligned}\mathcal{C}\left(\mathrm{Re}\left(\mathbf{Z}\right);\mathrm{Im}\left(\mathbf{Z}\right)\right)&=\mathbf{Z}\\ \mathcal{D}\left(\mathbf{Z}\right)&=\left(\mathrm{Re}\left(\mathbf{Z}\right);\mathrm{Im}\left(\mathbf{Z}\right)\right)\end{aligned}\tag{16}$$

where $\mathrm{Re}(\cdot)$ and $\mathrm{Im}(\cdot)$ represent the real and imaginary parts of the input matrix, respectively. $F^k(\cdot)$ and $\tilde{F}^k(\cdot)$ represent the transformations from the image domain to the sparse domain and back to the image domain, respectively, and they both use the simplest three-layer convolutional kernels with two intermediate rectified linear unit (ReLU) activations

$$F^k\left(x\right)=\mathbf{W}_{13}^k\otimes\delta\left(\mathbf{W}_{12}^k\otimes\delta\left(\mathbf{W}_{11}^k\otimes x+\mathbf{b}_{11}^k\right)+\mathbf{b}_{12}^k\right)+\mathbf{b}_{13}^k\tag{17}$$

$$\tilde{F}^k\left(x\right)=\mathbf{W}_{23}^k\otimes\delta\left(\mathbf{W}_{22}^k\otimes\delta\left(\mathbf{W}_{21}^k\otimes x+\mathbf{b}_{21}^k\right)+\mathbf{b}_{22}^k\right)+\mathbf{b}_{23}^k\tag{18}$$

where $\mathbf{W}_{ij}^k$ represents the weight of the jth 3×3 convolutional kernel in the ith module during the kth iteration, $\mathbf{b}_{ij}^k$ represents the corresponding bias, $\otimes$ denotes the convolutional operation, and δ represents the ReLU activation function. All the convolutions are implemented using group convolutions with a group size of 2, enabling the separate processing of the real and imaginary parts. Except for the last layer which has two filters, the output channels of other layers are fixed as $\mathcal{N}=64$. The learnable parameters in the **N** module include the thresholds $\{\mu^k\}_{k=1}^K$ and the convolutional layers $\{F^k(\cdot),\tilde{F}^k(\cdot)\}_{k=1}^K$, where K denotes the unrolling depth, i.e., the number of iterations of the optimization algorithm unfolded into trainable layers in the network.

In summary, the overall SAR imaging algorithm can be seen from Fig. 3. First, $\mathbf{X}^0$ is initialized as a zero matrix. The imaging algorithm consists of a total of K layers, where each layer defined by the **R** module in (14) and the **N** module defined in (15) is concatenated, and the final SAR imaging result is obtained from the output of the last layer, i.e., $\mathbf{X}^K$.

C. Joint Optimization Method of Learnable Pulse Transmission Pattern and SAR Imaging Strategy

Based on the learnable pulse transmission pattern in Section III-A and the SAR imaging method in Section III-B, we introduce a data-driven and end-to-end joint optimization method to improve SAR imaging quality, see Algorithm 1. The predefined SAR scenario $\mathbf{X}$ and SAR system parameters $\mathbf{\Theta}$ are taken as input, and Algorithm 1 will obtain the final imaging result $\mathbf{X}^K$ along with the learned optimal pulse transmission pattern and the matched SAR imaging algorithm parameters.

Specifically, in each iteration of Algorithm 1, the simulated azimuthal nonuniform echo and the SAR imaging algorithm are forward propagated to obtain the final imaging result, which is then compared with the predefined GT to compute mse loss as the objective of the joint optimization

$$\mathcal{L}\left(\mathbf{\Gamma}_{\mathrm{Joint}}\right)=\frac{1}{2J}\sum_{j=1}^{J}\left\|\mathbf{X}_j^K-\mathbf{GT}_j\right\|_2^2\tag{19}$$

where $\mathbf{X}_j^K$ and $\mathbf{GT}_j$ represents the final output image and GT of the jth sample, respectively.

Algorithm 1 Proposed JointSAR for High-Resolution and Wide-Swath SAR Imaging

Input: Simulation SAR scenario $\mathbf{X}$, determined SAR system parameters $\mathbf{\Theta}$, unrolling depth K

Output: Imaging results $\mathbf{X}^K$, learned parameters $\mathbf{\Gamma}_{\mathrm{Total}}=\{\mathbf{d}_\eta,\alpha^k,\mu^k,F^k(\cdot),\tilde{F}^k(\cdot)\}_{k=1}^K$,

Initialization: $\mathbf{\Gamma}_{\mathrm{Total}}$
while not converge **do**
  Generate simulated azimuth nonuniform echo: $\mathbf{Y}=mat(\mathbf{A}(\mathbf{T}_\eta,\mathbf{\Theta})vec(\mathbf{X}))+\mathbf{N}$
  Initialization: $\mathbf{X}^0\leftarrow\mathbf{0}$
  for $k=1$ to K **do**
    $\mathbf{Z}^k=\mathbf{X}^{k-1}+\alpha^k\cdot H(\mathbf{Y}-G(\mathbf{X}^{k-1}))$
    $\mathbf{X}^k=\mathbf{Z}^k+\mathcal{C}(\tilde{F}^k(E_{\mu^k}(F^k(\mathcal{D}(\mathbf{Z}^k)))))$
  end for
  Calculate loss function $\mathcal{L}(\mathbf{\Gamma}_{\mathrm{Net}})$
  Random parameter freezing: $\nabla_{\mathbf{d}_\eta}\mathcal{L}(\mathbf{\Gamma}_{\mathrm{Net}})\leftarrow\mathbf{m}\odot\nabla_{\mathbf{d}_\eta}\mathcal{L}(\mathbf{\Gamma}_{\mathrm{Net}})$
  Gradient descent update: $\mathbf{\Gamma}_{\mathrm{Net}}\leftarrow\mathbf{\Gamma}_{\mathrm{Net}}-\beta\nabla_{\mathbf{\Gamma}_{\mathrm{Net}}}\mathcal{L}(\mathbf{\Gamma}_{\mathrm{Net}})$
end while

In addition to the mse loss guided by the imaging results, we also introduce two regularization terms in the loss function, where one is for the pulse transmission pattern and the other for SAR imaging. Specifically, the regularization term for the pulse transmission pattern ensures the minimum pulse transmission interval required for wide-swath imaging. Each component in $\mathbf{d}_\eta$ must be constrained to be greater than 0, and the penalty term for the pulse transmission pattern is defined as follows:

$$\mathcal{L}\left(\mathbf{\Gamma}_{\mathrm{Transmission}}\right)=\sum_{n_a=1}^{\tilde{N}_a}\max\left(-d_{n_a},0\right).\tag{20}$$

To ensure the reversibility between the image domain and the sparse domain in the SAR imaging algorithm, $F^k(\cdot)$ and $\tilde{F}^k(\cdot)$ should be symmetric, i.e., $F^k(\cdot)\circ\tilde{F}^k(\cdot)=I$, where $\circ$ denotes the composition of functions, and $\tilde{F}^k(F^k(\mathbf{X}))=\mathbf{X}$. The penalty term of the SAR imaging part in the loss function is defined as follows:

$$\mathcal{L}\left(\mathbf{\Gamma}_{\mathrm{Imaging}}\right)=\frac{1}{2J}\sum_{j=1}^{J}\sum_{k=1}^{K}\left\|\tilde{F}^k\left(F^k\left(\mathbf{Z}_j^k\right)\right)-\mathbf{Z}_j^k\right\|_2^2\tag{21}$$

where J is the total number of samples.

In summary, the total loss function for the proposed JointSAR is defined as follows:

$$\begin{aligned}\mathcal{L}\left(\mathbf{\Gamma}_{\mathrm{Total}}\right)&=\gamma_1\mathcal{L}\left(\mathbf{\Gamma}_{\mathrm{Transmission}}\right)+\gamma_2\mathcal{L}\left(\mathbf{\Gamma}_{\mathrm{Imaging}}\right)+\gamma_3\mathcal{L}\left(\mathbf{\Gamma}_{\mathrm{Joint}}\right)\\&=\gamma_1\cdot\sum_{n_a=1}^{\tilde{N}_a}\max\left(-\mathbf{d}_{n_a},0\right)\\&\quad+\gamma_2\cdot\frac{1}{2J}\sum_{j=1}^{J}\sum_{k=1}^{K}\left\|\tilde{F}^k\left(F^k\left(\mathbf{Z}_j^k\right)\right)-\mathbf{Z}_j^k\right\|_2^2\\&\quad+\gamma_3\cdot\frac{1}{2J}\sum_{j=1}^{J}\left\|\mathbf{X}_j^K-\mathbf{GT}_j\right\|_2^2\end{aligned}\tag{22}$$

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where γ_1–γ_3 represent the weights of $\mathcal{L}(\mathbf{\Gamma}_{\text{Transmission}})$, $\mathcal{L}(\mathbf{\Gamma}_{\text{Imaging}})$, and $\mathcal{L}(\mathbf{\Gamma}_{\text{Joint}})$, respectively.

The learnable parameters in proposed method are denoted by $\mathbf{\Gamma}_{\text{Total}} = \{\mathbf{d}_\eta, \alpha^k, \mu^k, F^k(\cdot), \tilde{F}^k(\cdot)\}_{k=1}^K$ and parameters are trained without sharing during the K iterations. All the parameters are optimized using a mini-batch gradient descent method [57] based on the loss function defined in (22).

Furthermore, during the backpropagation optimization of the pulse transmission sequence, the dependency between adjacent elements in the sequence can cause the gradient to gradually diminish during propagation, hindering effective parameter updates and thus affecting the optimization performance. This may lead to the gradient directions of adjacent pulse transmission intervals becoming similar, lacking randomness, and thereby reducing the antialiasing capability of nonuniform pulse transmission pattern. To solve this problem, we propose a strategy of random parameter freezing to assist learning of the pulse transmission pattern. By randomly setting the gradients of the learnable parameters of the transmission sequence to 0 with a certain probability before parameter updates, a certain degree of randomness is introduced during the parameter gradient descent update process. The gradient of the loss function $\mathcal{L}(\mathbf{\Gamma}_{\text{Total}})$ in (22) with respect to the pulse transmission timing layer during backpropagation is represented as follows:

$$\nabla_{\mathbf{d}_\eta}\mathcal{L} = \left(\frac{\partial \mathcal{L}}{\partial d_1}, \ldots, \frac{\partial \mathcal{L}}{\partial d_{n_a}}, \ldots, \frac{\partial \mathcal{L}}{\partial d_{\tilde{N}_a}}\right). \tag{23}$$

We introduce a random mask vector $\mathbf{m} \in \mathbb{R}^{\tilde{N}_a}$, where each element m_{n_a} in $\mathbf{m}$ is 0 or 1, indicating whether to retain the gradient of the corresponding parameter. Specifically,

$$\mathbf{m} = \left(m_1, \ldots, m_{n_a}, \ldots, m_{\tilde{N}_a}\right), \quad m_{n_a} \sim \text{Bernoulli}\,(p) \tag{24}$$

where the freezing probability factor p is the probability of preserving gradients. Then, for the pulse transmission parameter $\mathbf{d}_\eta$, its gradient is revised to

$$\tilde{\nabla}_{\mathbf{d}_\eta}\mathcal{L} = \mathbf{m} \odot \nabla_{\mathbf{d}_\eta}\mathcal{L} \tag{25}$$

where $\odot$ means the Hadamard product.

This strategy randomly freezes the gradients of certain parameters during each iteration, similar to Dropout [58], but operates on gradients rather than neuron activation values in hidden layers. In this task, we set the pulse transmission interval as the neuron of the input layer for training. If we randomly set it to 0 each iteration, like Dropout, it will greatly affect and make the pulse transmission pattern obtained each iteration different, affecting the stability of training. By applying random freezing to gradients, our method can introduce randomness without affecting the training stability, making it more effective than Dropout in hidden layers in the task of input layer sequence generation. In Section IV-D, we provide a detailed discussion on the differences between various learning methods for the parameters $\mathbf{d}_\eta$.

D. Analysis on Parameter Quantity and Computational Complexity

We analyze the number of parameters in the training procedure and the computational complexity in the inference process of our proposed JointSAR. Among the total learnable parameters $\mathbf{\Gamma}_{\text{Total}} = \{\mathbf{d}_\eta, \alpha^k, \mu^k, F^k(\cdot), \tilde{F}^k(\cdot)\}_{k=1}^K$, $\mathbf{d}_\eta$ contains $\tilde{N}_a$ parameters, while $F^k(\cdot)$ and $\tilde{F}^k(\cdot)$ consist of six group convolutions with $\mathcal{N}$ intermediate channels and w_f kernel size. The number of parameters for $F^k(\cdot)$ or $\tilde{F}^k(\cdot)$ is calculated as $2w_f w_f \mathcal{N}/2 + 2\mathcal{N} w_f w_f \mathcal{N}/2 = \mathcal{N}(1+\mathcal{N})w_f^2$. Consequently, the overall number of parameters in the training procedure is $\tilde{N}_a + K(2 + 2\mathcal{N}(1+\mathcal{N})w_f^2)$.

In practical applications, we transmit pulses based on predefined azimuthal transmission timings, obtain the echoes, and use the pretrained SAR imaging model to generate wide-swath images. Next, we focus on the computational complexity of the SAR imaging part after obtaining the echoes in the inference procedure of our proposed method. In the linear **R** module, the primary components are the SAR imaging operator $H(\cdot)$ and the echo simulation operator $G(\cdot)$. Each operator involves one matrix multiplication, three matrix pointwise multiplications, and three FFT operations, so the total computational cost is $2N_aN_r(\tilde{N}_a + 3 + \log N_a + 2\log N_r)$. In the nonlinear **N** module, there are six group convolutions with the total computational cost given by $2(2\mathcal{N}+2\mathcal{N}^2)(w_f^2 N_a N_r)/2 = 2N_a N_r w_f^2 \mathcal{N}(\mathcal{N}+1)$. Thus, the overall computational complexity of the imaging algorithm is $K(2N_aN_r(\tilde{N}_a + 3 + \log N_a + 2\log N_r + w_f^2\mathcal{N}(\mathcal{N}+1)))$. From the aforementioned analysis, the overall computational complexity of SAR imaging process is mainly determined by two parts, one is the **R** module associated with the convolution, and the other is the **N** module highly related to the size of imaging scene $N_a \times N_r$ and the number of transmission pulses $\tilde{N}_a$. When the size of the echo is small, the computational cost is primarily driven by the **N** module. As the size of the imaging scene and measured echo significantly increases, the computational cost associated with the convolutional part varies little. Instead, the main computational bottleneck shifts to the number of transmitted pulses $\tilde{N}_a$ in the case of a large imaging scene.

IV. Experimental Results

We compare the proposed JointSAR with the classical and the state-of-the-art competing SAR imaging methods through simulated and measured SAR datasets in terms of both visual effects and evaluation indicators. In addition, the performance of the proposed method is further demonstrated by experiments regarding the sidelobe suppression analysis, point target aliasing effect analysis, ablation studies of algorithm modules, and universality analysis.

A. Experimental Setup

This section introduces our dataset, parameter settings, competing methods, and evaluation indicators.

1) Datasets: In this article, we use simulated data for training and inference, and the measured data are only used for inference. During training, we preset the SAR scenarios of point targets without generating the echoes in advance, as the echoes change with the pulse transmission timing. A total of 10 000 simulated scenarios are setup, which are divided into training, validation, and testing in a ratio of 70%, 20%, and 10%. Radar parameters for the simulation are set according to the Simulation and RadarSAT-1 parameters shown in Table II. The simulation parameters are used to verify the ability to achieve wide-swath and suppress range ambiguity, while the

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TABLE II
SAR SYSTEM PARAMETERS

Parameters	Simulation	RadarSAT-1	Gaofen-3 (Data1)	Gaofen-3 (Data2)	Taijing-4	AS01
Bandwidth: $B(\mathrm{MHz})$	2	30.111	60	40	40	150
Radar center frequency: $F_c(\mathrm{GHz})$	9	5.3	5.404	5.400	15.3	9.6
Radar sampling frequency: $F_s(\mathrm{MHz})$	2.7	32.317	66.66	66.67	50	200
Equivalent radar velocity: $V(\mathrm{m/s})$	7100	7062	7126.06	7151.86	7348.6	7446.4
Pulse duration: $T_p(\mu\mathrm{s})$	30	41.74	25	24.99	30	18.4
Pulse repetition frequency: $\mathrm{PRF(Hz)}$	6000	1256.98	2589.17	2435.20	7000	5433.6
Slant range of scene center: $R_s(\mathrm{km})$	518.75	994.21	899	908.45	568.15	537.01
Beam squint angle: $\theta\,(\mathrm{rad})$	0	-0.0277	0	0	0	0

RadarSAT-1 parameters are used to validate its high-resolution imaging capability.

Measured data are used for testing to validate the imaging performance of different methods. Since we cannot directly acquire echoes from specific transmission timings, we generate azimuthal nonuniform echo by interpolating the received real echoes based on the pulse transmission timings of different methods, as operated in [38] and [40]. Specifically, we employ cubic interpolation along the azimuth dimension to generate the nonuniform echoes, where the original uniformly sampled azimuth positions are treated as input nodes and the desired nonuniform sampling timings are used as query points for interpolation. Notably, the measured data are not involved in model training. We used measured data from multiple satellites involving RadarSAT-1, Gaofen-3, Taijing-4, and AS01 satellites. The SAR system parameters for each scenario are shown in Table II. More details regarding these datasets are introduced below.

RadarSAT-1, a well-known spaceborne SAR satellite launched in 1995, was the source of the raw data used in this article, which was collected on June 16, 2002. For this experiment, we selected two typical scenes from RadarSAT-1 data, featuring several ships distributed on the sea surface.

Gaofen-3 satellite, Chinese C-band multipolarization SAR satellite, was launched in August 2016. In this experiment, we selected two SAR datasets (Data1 and Data2) captured at different times by Gaofen-3, with their parameters detailed in Table II.

The Taijing-4 satellite, recently launched by MinoSpace, provided the azimuthally nonuniform dataset used in this article. This dataset was transmitted using nonuniform pulse transmission patterns, i.e., the random PRF variation method in [40].

The AS01 satellite, launched by Skysight Technology, is equipped with a lightweight SAR payload based on a 2-D phased array antenna. In this experiment, we use the AS01 satellite dataset of a wide-swath coastal scene near Singapore to demonstrate the performance of the proposed method.

2) Hyperparameter Settings: For training, we set the original wide-swath and interval to just satisfy the Nyquist sampling theorem and the extended swath width to be 1.5 times the original wide-swath (the average downsampling rate of SAR echo to be 50%); that is, the average and minimum pulse transmission interval are 2 and 1.5 times the original interval, respectively.The unrolling depth of the proposed unfolding SAR imaging algorithm, i.e., K, is set to 4, the learnable intervals $\mathbf{d}_\eta$ are initialized as uniformly distributed random variables, and the freezing probability factor p of the random parameter freezing is set to 0.7. Parameters α^k and μ^k are initialized to 0.01 and 0.02, respectively. Weights in the CNNs layers $F^k(\cdot)$ and $\tilde{F}^k(\cdot)$ are initialized using the Glorot normal distribution method [59], and the biases are initialized to 0. γ_1, γ_2 and γ_3 in loss function is set as 1, 0.01, and 1, respectively. We set 10 epochs for training with a learning rate of 0.001, and the Adam [60] optimizer is used to minimize the loss function. Our method is trained in Python 3.11.5 with Pytorch 2.0.0 on a Linux system with an NVIDIA A800 80G.

3) Competing Methods: To compare the effectiveness of the proposed JointSAR, we selected several classical and state-of-the-art high-resolution and wide-swath imaging models. First, we select CSA, a typical matched filter imaging method to process full echoes that meet the Nyquist sampling frequency as a baseline reference (full CSA). Then, different pulse transmission patterns and corresponding SAR imaging methods are introduced for comparison. Specifically, we compare the traditional matched filter method (uniform CSA) and the ISTA imaging algorithm (uniform ISTA), respectively, with a uniform downsampling pulse transmission pattern. To maintain a consistent data volume, the average downsampling rate was set to 50%, which allows for a twofold bandwidth expansion in the case of uniform downsampling. For nonuniform pulse transmission patterns, we compare the Poisson disk transmission pattern with the ISTA imaging method [38] (Poisson disk ISTA) and the random PRF variation transmission pattern with the fast ISTA imaging method [40] (R-V Omega-K FISTA). Similarly, to ensure a fair comparison, the downsampling rate and wide-swath expansion factor for nonuniform transmission patterns were set to match those of the proposed method, achieving 1.5 times wide-swath expansion with 50% of the data volume.

DUN-based SAR imaging use random identity matrix or random gaussian matrix [12], [13], [14], [44], [53], [54], [61] for echo sampling, corresponding to the random sampling selection pulse transmission pattern, which cannot achieve a wide-swath. We select DUN-AO-Net [14] as the representative work for comparison due to its good imaging performance.

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4) Evaluation Indicators: To assess the performance of different methods, a combination of multiple evaluation metrics [45], [46], [54], [62], and visual analysis is employed. Normalized mean-square error (NMSE) is used to evaluate the reconstruction error between reconstructed image and GT. The expression for NMSE is as follows:

$$\text{NMSE} = \frac{\|\mathbf{X}_T - \mathbf{X}_L\|_2^2}{\|\mathbf{X}_L\|_2^2} \tag{26}$$

where $\mathbf{X}_T$ is the reconstructed image and $\mathbf{X}_L$ is the predefined GT. Peak signal-to-noise ratio (PSNR) is used to measure the signal-to-noise ratio (SNR) between reconstructed image and GT

$$\text{PSNR} = 20 \times \lg\left(\frac{\max(\mathbf{X}_L)}{\sqrt{\frac{1}{N}\|\mathbf{X}_T - \mathbf{X}_L\|_2^2}}\right) \tag{27}$$

where $\max(\mathbf{X}_L)$ is the maximum possible pixel value of GT and N is the total number of pixels in the image. Target-to-background ratio (TBR) is used to measure the contrast between the target area and the background area. The expression is as follows:

$$\text{TBR} = 20 \times \log_{10}\left(\frac{\|\mathbf{X}_T \in \Omega_T\|_2^2}{\|\mathbf{X}_T \in \Omega_B\|_2^2}\right) \tag{28}$$

where Ω_T represents the target region and Ω_B represents the background region. Image entropy (ENT) measures the average information and sparsity of the image, with lower values indicating a sparser, more refined, and focused imaging result

$$\text{ENT} = -\left\|\frac{\mathbf{X}_T}{g}\right\|_2^2 \ln\left\|\frac{\mathbf{X}_T}{g}\right\|_2^2 \tag{29}$$

where $g = \|\mathbf{X}_T\|_2^2$ is the normalization constant. For real-world scenes without GT, NMSE, and PSNR metrics are infeasible. In contrast, TBR requires only pixel areas containing targets, and ENT does not rely on GT, making TBR and ENT more suitable for using the measured data.

In addition, the peak sidelobe ratio (PSLR) and the integrated sidelobe ratio (ISLR) [40] were employed to assess the sidelobe levels in the point target test. PSLR measures the ratio between the highest sidelobe and the peak of the main lobe in the image

$$\text{PSLR} = 20 \times \log_{10}\left(\frac{\max(\|\mathbf{X}_T \in \Omega_S\|)}{\max(\|\mathbf{X}_T \in \Omega_M\|)}\right) \tag{30}$$

where Ω_M represents the main lobe region and Ω_S denotes the sidelobe region. ISLR measures the ratio of the total energy in the sidelobes to the total energy in the main lobe

$$\text{ISLR} = 20 \times \log_{10}\left(\frac{\sum_{\Omega_S}\|\mathbf{X}_T\|}{\sum_{\Omega_M}\|\mathbf{X}_T\|}\right). \tag{31}$$

A lower PSLR and ISLR suggest enhanced main lobe energy and reduced sidelobe energy, indicating the improved effectiveness.

B. Experiments on Simulated Data

First, we verify the effectiveness of the proposed method in improving imaging quality and extending the maximum unambiguous detection swath under wide-swath and range ambiguity scenarios. The simulation parameters are set in Table II. According to the analysis and calculation in Section II-C and (11), the achievable swath width under these parameters in Table II is 16 km. We design two simulated scenes with different coverage widths which exceed the maximum unambiguous detection swath of 16 km. The imaging scenes contain nine targets arranged in a 3×3 grid in azimuth and range directions. In the two scenes, the targets cover ranges of 17 and 22 km, respectively, with their exact positions marked in Fig. 4. For the 17-km wide-swath scenario, the echoes from the third, sixth, and ninth targets return to the radar during the transmission period in the single-channel SAR mode under a full-pulse transmission pattern, when the receiver is inactive. As a result, it is impossible to obtain complete target information using the full-pulse transmission pattern without expanding the swath width. In the 22-km wide-swath scenario, the echoes from the third, sixth, and ninth targets overlap with the first, fourth, and seventh targets' echoes during the subsequent reception time window, leading to a range ambiguity problem. By increasing the minimum pulse transmission interval, the proposed method achieves a larger unambiguous swath width and effectively improves the imaging quality.

Fig. 5 illustrates the imaging results of the two scenes. The first and second rows in Fig. 5 show the imaging results of the 17- and 22-km coverage scenes using different methods, respectively. As shown in Fig. 5(a), the full-CSA method fails to achieve accurate reconstruction in wide swath and range ambiguity scenarios without swath width expansion. Fig. 5(b) and (c) employs a uniform downsampling pulse transmission pattern, enabling a twofold swath width expansion. However, azimuth aliasing occurs due to the sub-Nyquist sampling and is difficult to eliminate through ISTA, as shown in Fig. 5(c). Fig. 5(d) and (e) adopts nonuniform pulse transmission patterns, achieving a 1.5 times swath width expansion while converting aliasing into noise-like artifacts for mitigation. However, the overall imaging quality remains to be further improve. Fig. 5(f) applies a DUN-based SAR imaging algorithm to improve the imaging quality but utilizes a random selection pulse transmission pattern, which prevents the swath width expansion and leads to incomplete imaging. The proposed method jointly optimizes the learnable pulse transmission pattern and imaging algorithm, achieving a 1.5 times swath width expansion to obtain complete target information while significantly improving SAR imaging quality, leading to superior reconstruction results in Fig. 5(g). Table III shows the average evaluation indicators for the two scenes in Fig. 5. The proposed method achieves lower NMSE and higher PSNR with swath width expansion, while obtaining higher TBR with lower ENT, indicating better target imaging performance.

Furthermore, to verify the imaging quality of the proposed method under more realistic squint angle conditions, we conducted simulation experiments using RadarSAT-1 parameters. In this experiment, the scene size was set to 512×512, with randomly generated point targets and elliptical targets as labels.To clearly compare the effectiveness of different meth-

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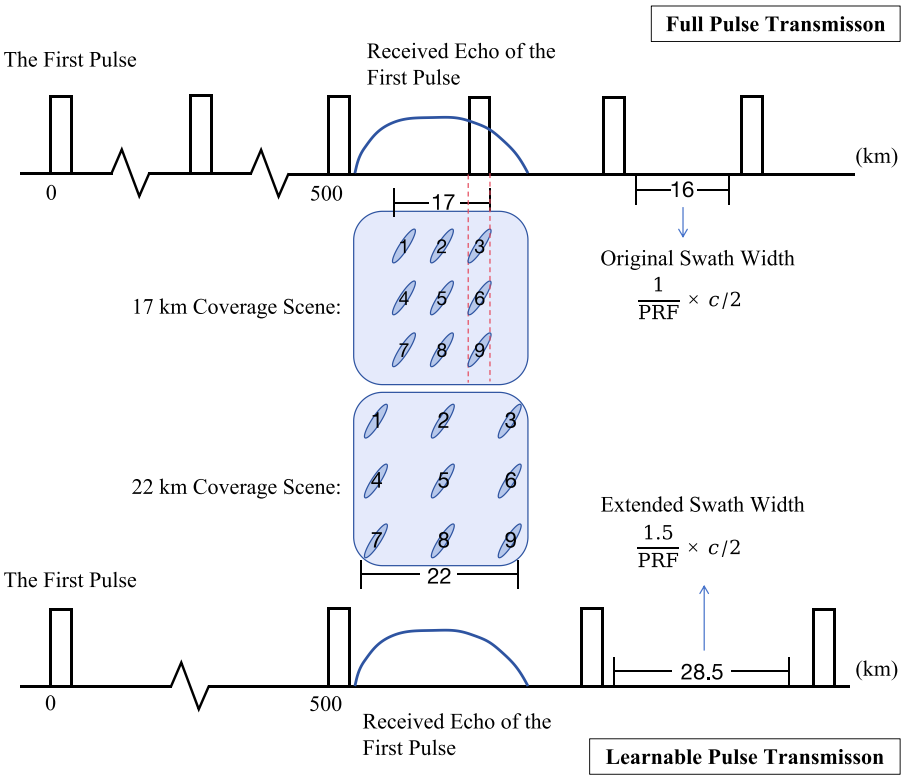

Fig. 4. Scene settings for validating wide-swath imaging and range ambiguity suppression. The upper part shows how the wide-swath scenario in the full-pulse transmission pattern leads to echo conflicts. The lower part demonstrates how the proposed method, by increasing the minimum pulse transmission interval, achieves a wider swath while suppressing range ambiguity.

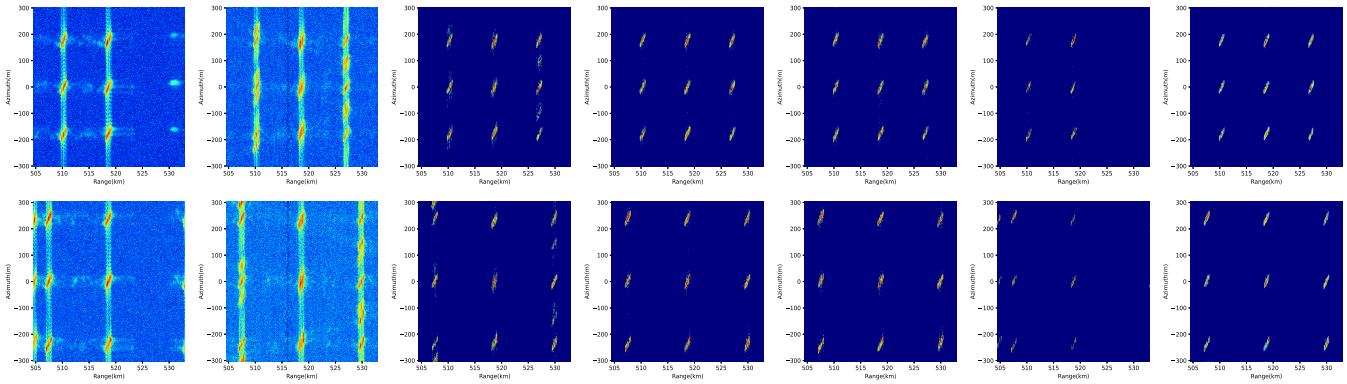

Fig. 5. Visual comparison of imaging results of different methods for wide-swath and range ambiguity suppression under simulation parameters. The first row corresponds to the 17-km coverage scene for validating the wide-swath scene, while the second row corresponds to the 22-km coverage scene for validating the suppression of range ambiguity. More details on scene settings can be found from Fig. 4. (a) Full CSA, (b) uniform CSA, (c) uniform ISTA, (d) Poisson disk ISTA [38], (e) R-V Omega-K FISTA [40], (f) DUN-AO-Net [14], and (g) proposed method.

ods, we selected a simulated image with 30 point targets and another with nine elliptical targets. Fig. 6 illustrates the imaging results of various methods, with point targets and elliptical targets shown in two respective rows. The results reveal that a uniform downsampling pulse transmission pattern leads to obvious aliasing, which cannot be entirely reduced even with compressive sensing-based methods. In contrast, other nonuniform transmission patterns provide relatively better imaging quality. Since DUN-AO-Net [14] utilizes the simple RDA, it struggles to achieve proper focus in scenarios involving a squint angle, leading to poorer resolution. Conversely, the proposed method demonstrates superior performance in target focusing. The evaluation metrics for these two images are provided in Tables IV and V, showing that our method achieves the better imaging performance than the compared methods.

As an additional comparison to illustrate the advantages of the proposed method over conventional iterative approaches, Fig. 7 presents the imaging results of the Poisson disk ISTA method under different sparsity thresholds. It can be observed

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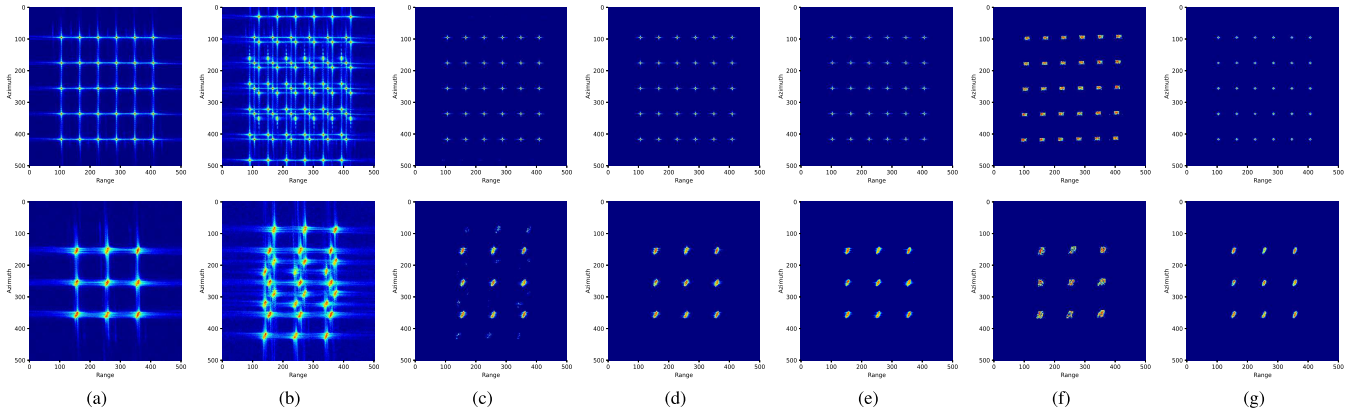

Fig. 6. Visual comparison of imaging results of different methods for point targets and elliptical targets scenes. The first row is a point target scene, and the second row is an elliptical target scene. (a) Full CSA, (b) uniform CSA, (c) uniform ISTA, (d) Poisson disk ISTA [38], (e) R-V Omega-K FISTA [40], (f) DUN-AO-Net [14], and (g) proposed method.

TABLE III

AVERAGE EVALUATION INDICATORS FOR THE TWO NINE-ELLIPSE SIMULATED SCENARIOS, VALIDATING WIDE SWATH AND RANGE AMBIGUITY SUPPRESSION UNDER SIMULATION PARAMETERS. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	NMSE	PSNR	TBR	ENT
Full CSA	1.63	28.86	-4.75	8.75
Uniform CSA	2.38	27.19	-10.77	9.49
Uniform ISTA	0.91	31.33	0.90	7.14
PoissonDisk ISTA [38]	0.83	31.71	3.68	7.11
R-V Omega-K FISTA [40]	0.93	31.26	0.71	7.09
DUN-AO-Net [14]	0.93	31.21	1.92	**6.07**
Proposed	**0.78**	**32.01**	**6.00**	6.78
Ideal value	**0**	∞	∞	**0**

TABLE IV

EVALUATION INDICATORS ON A SIMULATE SCENARIO WITH 30 POINT TARGETS. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	NMSE	PSNR	TBR	ENT
Full CSA	6.12	31.53	8.00	6.77
Uniform CSA	10.94	29.02	0.93	8.02
Uniform ISTA	5.19	32.25	10.21	5.93
Poisson disk ISTA [38]	5.06	32.36	10.43	5.91
R-V Omega-K FISTA [40]	4.88	32.52	9.75	5.86
DUN-AO-Net [14]	13.82	28.00	-26.13	7.64
Proposed	**4.07**	**33.30**	**12.78**	**5.53**
Ideal value	**0**	∞	∞	**0**

that although increasing the threshold suppresses sidelobes to some extent, it also leads to degraded mainlobe intensity and blurred target edges. This indicates that while higher thresholds reduce sidelobe levels, they compromise the mainlobe reconstruction, making it difficult to achieve both strong sidelobe suppression and clear target reconstruction simultaneously. In contrast, the proposed method effectively suppresses sidelobes while maintaining high mainlobe intensity and sharp target boundaries, resulting in superior overall imaging quality.

To further compare the sidelobe suppression capabilities of different methods, the first row of Fig. 8 displays an

TABLE V

EVALUATION INDICATORS ON A NINE-ELLIPSE SIMULATED SCENARIO. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	NMSE	PSNR	TBR	ENT
Full CSA	1.99	28.65	9.21	7.36
Uniform CSA	3.04	26.82	-4.57	8.46
Uniform ISTA	1.68	29.39	13.87	6.69
PoissonDisk ISTA [38]	1.75	29.20	14.49	6.68
R-V Omega-K FISTA [40]	1.66	29.43	14.37	6.66
DUN-AO-Net [14]	1.91	28.82	-3.47	6.92
Proposed	**1.51**	**29.85**	**20.32**	**6.32**
Ideal value	**0**	∞	∞	**0**

TABLE VI

PSLR AND ISLR OF DIFFERENT METHODS IN FIG. 8 SCENE. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	PSLR(dB)	ISLR(dB)
Full CSA	-13.11	5.33
Uniform CSA	-9.26	15.59
Poisson disk ISTA [38]	-14.17	-7.40
Proposed	**-21.05**	**-16.50**
Ideal value	$-\infty$	$-\infty$

intensity heatmap of the imaging results within a 50×50 area around a single point target, providing a clearer view of sidelobe suppression in both azimuth and range dimensions. The second row of Fig. 8 illustrates the amplitude distribution in the azimuth dimension, demonstrating that our method is more effective in sidelobe suppression while maintaining a narrow main lobe width in comparison with other methods. In addition, Table VI presents the PSLR and ISLR metrics, where the proposed method achieves the lowest values, indicating superior main lobe identification and imaging quality of our JointSAR.

To compare the structural aliasing and noise-like artifacts caused by nonuniform pulse transmission before and after training, Fig. 9 represents the spectral aliasing resulting from the initialized nonuniform transmission pattern before training, the optimized transmission pattern after only training of the

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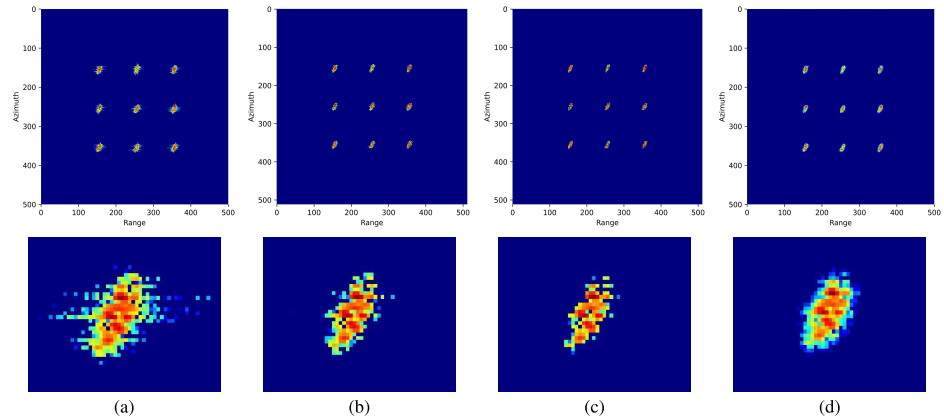

Fig. 7. Visual comparison of imaging results for scenes with nine elliptical targets using the Poisson disk ISTA method with different sparsity thresholds and the proposed method. (a) Poisson disk ISTA (threshold = 1), (b) Poisson disk ISTA (threshold = 2), (c) Poisson disk ISTA (threshold = 3), and (d) proposed method.

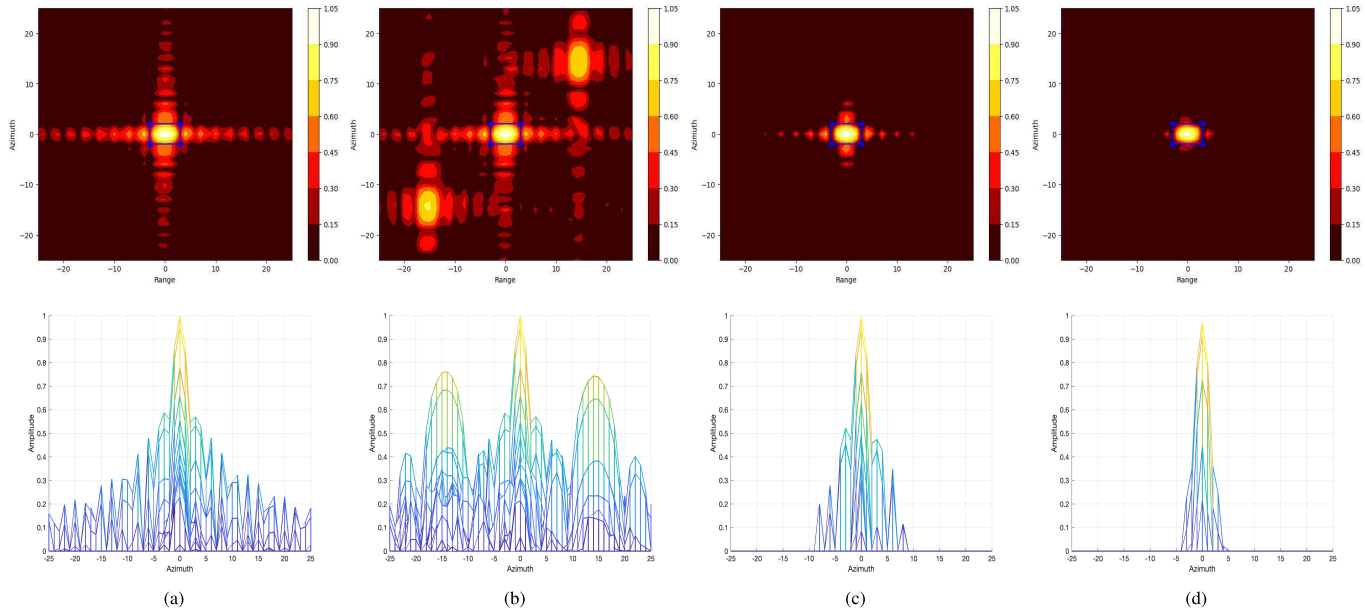

Fig. 8. Comparison of sidelobe suppression effects for a single point target. The first row is a 2-D intensity heatmap representation, and the second row is the amplitude comparison in the azimuth direction. (a) Full CSA, (b) uniform CSA, (c) Poisson disk ISTA [38], and (d) proposed method.

pulse transmission pattern, and the joint optimization of the proposed JointSAR, respectively. It is clear that the maximum spectral aliasing caused by the nonuniform transmission pattern is significantly reduced after training. Furthermore, our proposed joint optimization strategy of pulse transmission pattern and SAR imaging method can greatly reduce the aliasing introduced by the pulse transmission pattern with downsampling.

C. Experiments on Measured Data

In addition to simulated data, we also use measured data to compare the performance of different methods. Furthermore, to evaluate the feasibility of wide-swath imaging, we used a dataset from the AS01 satellite over a coastal scene.

1) RadarSAT-1 Data: Fig. 10 presents the reconstruction results of different methods, demonstrating that the proposed method provides better focusing and effective aliasing suppression. Table VII provides a comparison of TBR and ENT metrics across different methods, with the proposed method achieving the highest TBR and lowest ENT, indicating superior ship focusing and clutter reduction. For a more detailed comparison, Fig. 11 presents the 3-D imaging details of the

TABLE VII

AVERAGE EVALUATION INDICATORS ON TWO REAL-WORLD RADARSAT-1 SAR EXAMPLES. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	TBR	ENT
Full CSA	3.49	4.93
Uniform CSA	-8.71	9.15
Uniform ISTA	23.14	5.09
Poisson disk ISTA [38]	<u>36.39</u>	<u>4.91</u>
R-V Omega-K FISTA [40]	33.41	4.93
DUN-AO-Net [14]	-7.08	6.37
Proposed	**47.49**	**4.06**
Ideal value	∞	**1**

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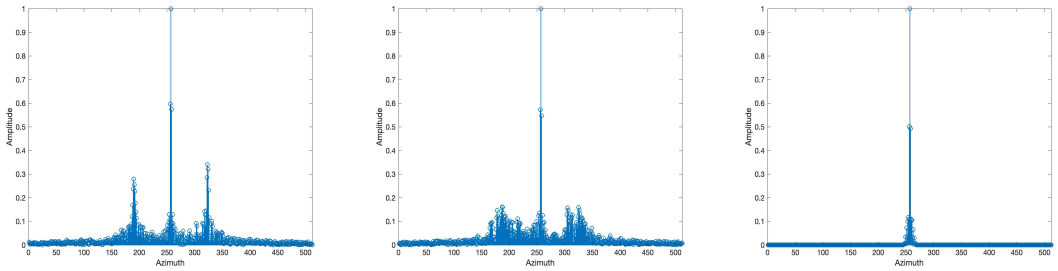
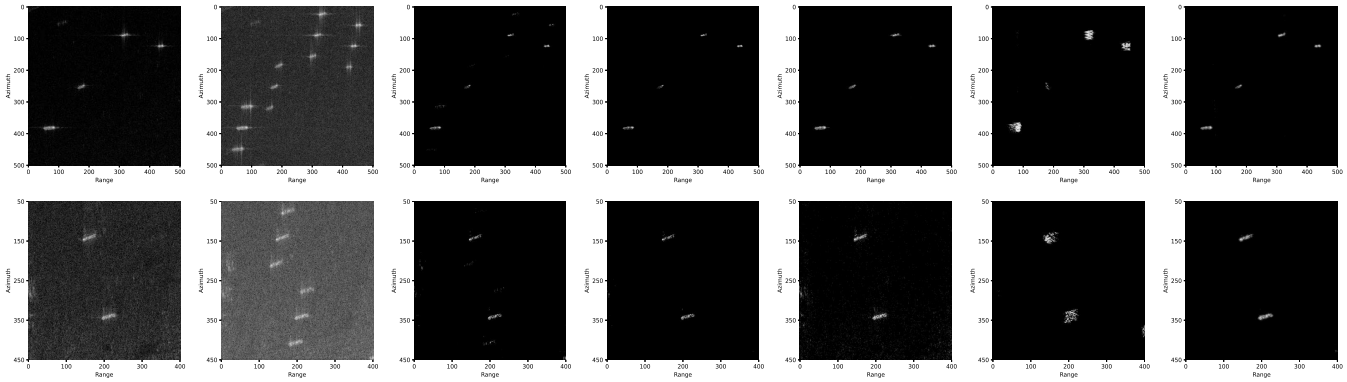


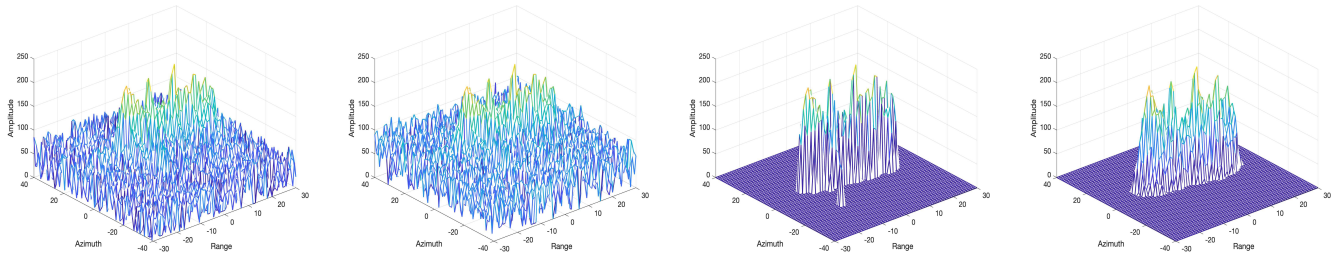
(a) (b) (c)

Fig. 9. Comparison of azimuth aliasing suppression capability of nonuniform transmission timings before and after training. (a) Aliasing caused by initial transmission timings before training, (b) aliasing caused by transmission timings after only training of pulse transmission pattern, and (c) aliasing suppression after our joint optimization of pulse transmission pattern and SAR imaging method.



(a) (b) (c) (d) (e) (f) (g)

Fig. 10. Visual comparisons of imaging results obtained by different methods on the RadarSAT-1 dataset. (a) Full CSA, (b) uniform CSA, (c) uniform ISTA, (d) Poisson disk ISTA [38], (e) R-V Omega-K FISTA [40], (f) DUN-AO-Net [14], and (g) proposed method.



(a) (b) (c) (d)

Fig. 11. Detailed imaging results of the second ship in the second image from the top in Fig. 10 from the RadarSAT-1 dataset. (a) Full CSA, (b) uniform CSA, (c) Poisson disk ISTA [38], and (d) proposed method.

second ship from the second image in Fig. 10. The results in Fig. 10 clearly demonstrate that the proposed method exhibits fewer sidelobes and superior focusing performance.

2) Gaofen-3 Data: Fig. 12 presents three selected subscenes from Data1. Due to the data acquisition of Gaofen-3 satellite at a squint angle of 0, DUN-AO-Net [14] performs better in this scenario than it does on the RadarSAT-1 dataset. Table VIII presents the average evaluation metrics for eight selected subscenes, where the proposed method achieves the highest TBR and the second-best ENT values. For Data2, a scene with a moving ship was selected. Fig. 13 and Table IX, respectively, display the imaging results and comparison metrics for the ships in Gaofen-3 Data2. In this dataset, the proposed method achieves the best ship focusing performance with the highest TBR. For Data2, a scene with a moving ship was selected. Fig. 13 and Table IX, respectively, display the imaging results and comparison metrics for the ships in Gaofen-3 Data2. In this dataset, the proposed method achieves better ship imaging performance. It should be noted that TBR is more important than ENT when the primary criterion is the minimization of missed detections and the improvement of detection accuracy.

3) Taijing-4 Data: Based on the imaging parameters provided in Table II, it can be calculated that the azimuth resolution is nearly four times the range resolution, resulting in ships appearing slightly elongated in the azimuth direction. Fig. 14 and Table X show the imaging results and evaluation metrics for this dataset. Similarly, the proposed

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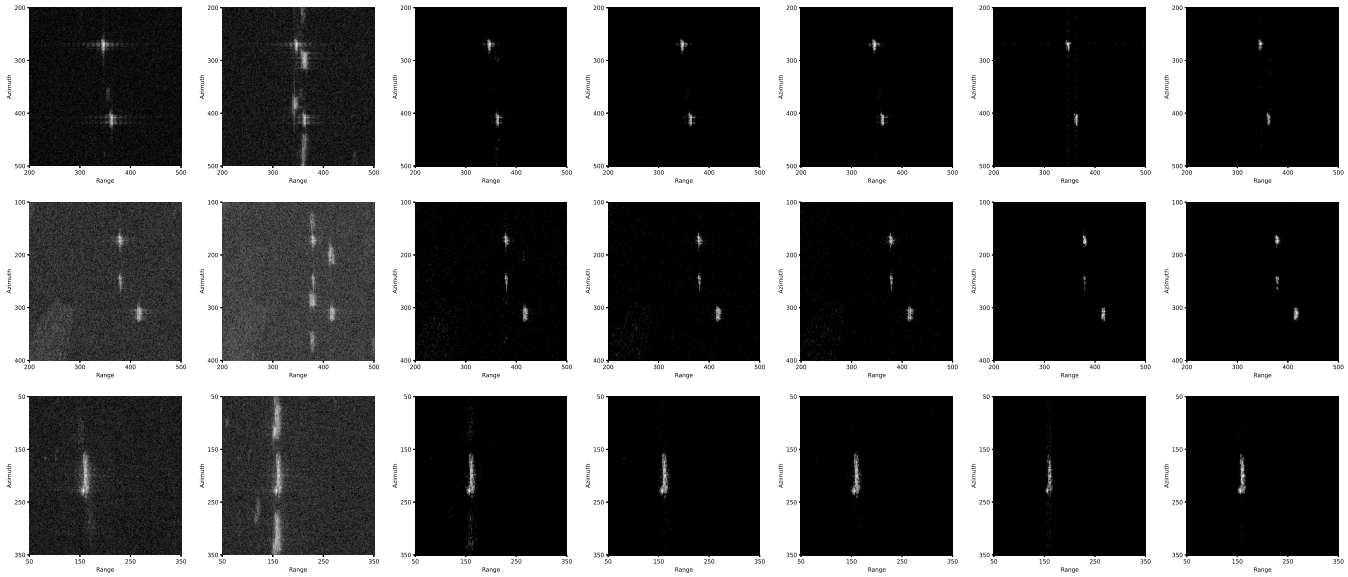
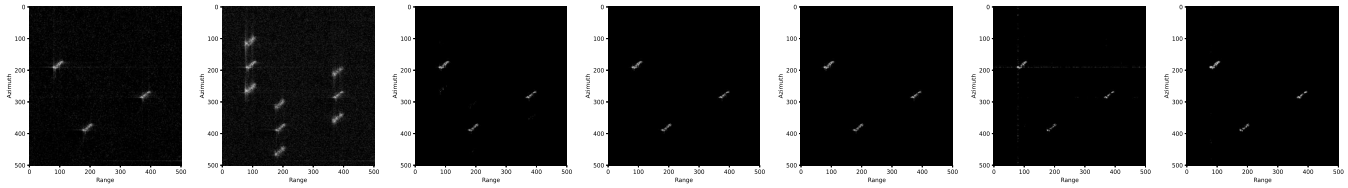


(a) (b) (c) (d) (e) (f) (g)

Fig. 12. Visual comparisons of imaging results obtained by different methods on Gaofen-3 Data1. (a) Full CSA, (b) uniform CSA, (c) uniform ISTA, (d) Poisson disk ISTA [38], (e) R-V Omega-K FISTA [40], (f) DUN-AO-Net [14], and (g) proposed method.



(a) (b) (c) (d) (e) (f) (g)

Fig. 13. Visual comparisons of imaging results obtained by different methods on Gaofen-3 Data2. (a) Full CSA, (b) uniform CSA, (c) uniform ISTA, (d) Poisson disk ISTA [38], (e) R-V Omega-K FISTA [40], (f) DUN-AO-Net [14], and (g) proposed method.

TABLE VIII

AVERAGE EVALUATION INDICATORS ON EIGHT REAL-WORLD GAOFEN-3 DATA1 SAR EXAMPLES. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	TBR	ENT
Full CSA	-6.91	9.98
Uniform CSA	-15.51	10.59
Uniform ISTA	16.97	5.81
Poisson disk ISTA [38]	<u>18.88</u>	5.55
R-V Omega-K FISTA [40]	17.78	5.60
DUN-AO-Net [14]	18.63	**4.26**
Proposed	**22.10**	<u>4.60</u>
Ideal value	∞	**1**

TABLE IX

EVALUATION INDICATORS ON A REAL-WORLD SAR EXAMPLE FROM GAOFEN-3 DATA2. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	TBR	ENT
Full CSA	11.73	6.991
Uniform CSA	-3.60	8.325
Uniform ISTA	37.76	4.988
Poisson disk ISTA [38]	<u>40.61</u>	5.001
R-V Omega-K FISTA [40]	40.29	4.989
DUN-AO-Net [14]	37.77	**4.375**
Proposed	**42.90**	<u>4.920</u>
Ideal value	∞	**1**

method demonstrates finer ship localization and improved visual clarity in the reconstructed images, while also achieving the higher TBR and comparable ENT values in comparison with other methods.

4) Wide-Swath Scenarios: In this section, we validate the feasibility of wide-swath imaging for large scenes and the generalization ability of the proposed method to input echoes with different sizes. The proposed method enables wide-swath imaging, defined as the capability to observe scenes at greater distances, either by extending the scene center range or increasing azimuthal coverage around it. To demonstrate this, we present the imaging results for a wide-swath coastal scene from the AS01 satellite, after only training on small-scene simulated data.

Fig. 15 shows the imaging result of wide-swath data from the AS01 satellite over a coastal scene near Singapore. The original echoes consist of 6000 azimuth pulses and 10 000 range samples. The method was retrained on 1024×1024 scenes using the AS01 radar parameters, resulting in around 512 learned azimuth pulse transmission intervals. For the larger scene in Fig. 15, the original 6000 azimuth pulses were downsampled to approximately 3000. To match this size,

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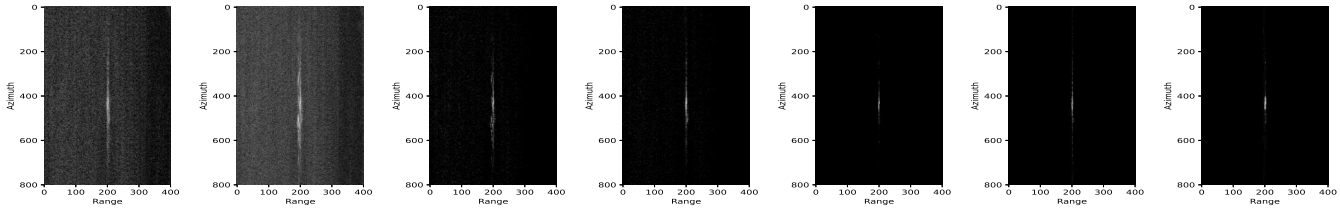

Fig. 14. Visual comparisons of imaging results obtained by different methods on nonuniform Taijing-4 dataset. (a) Full CSA, (b) uniform CSA, (c) uniform ISTA, (d) Poisson disk ISTA [38], (e) R-V Omega-K FISTA [40], (f) DUN-AO-Net [14], and (g) proposed method.

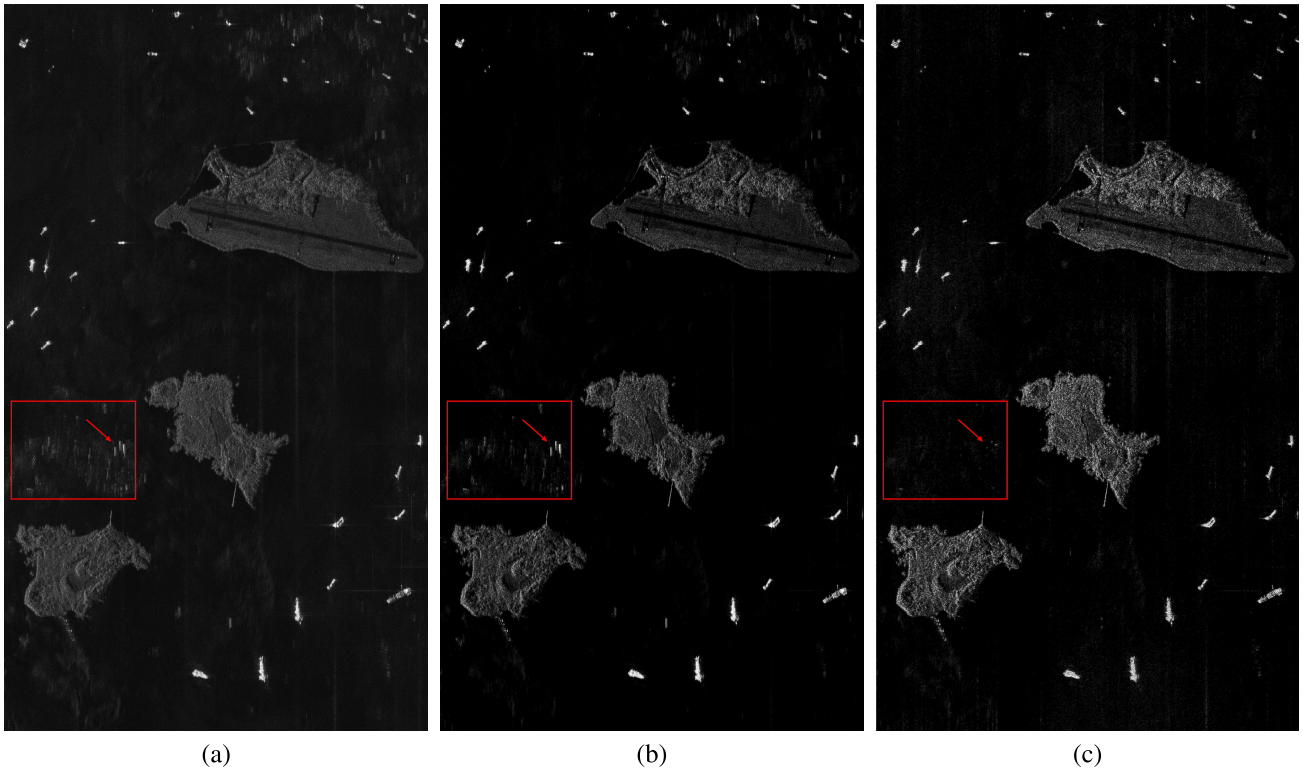

Fig. 15. Imaging results for the wide-swath coastal scene of the AS01 satellite. The red boxes and arrows indicate the regions where ghost images mainly occur. (a) Full CSA, (b) full ISTA, and (c) proposed method.

TABLE X
EVALUATION INDICATORS ON A REAL TAIJING-4 NONUNIFORM SAR EXAMPLE. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE SECOND PLACE IS UNDERLINED

Method	TBR	ENT
Full CSA	-11.70	11.56
Uniform CSA	-17.96	11.70
Uniform ISTA	9.22	6.60
Poisson disk ISTA [38]	25.46	5.89
R-V Omega-K FISTA [40]	25.53	**3.75**
DUN-AO-Net [14]	<u>28.18</u>	<u>4.64</u>
Proposed	**29.02**	4.74
Ideal value	∞	**1**

the learned pulse sequence of 512 transmission timings was periodically extended to 3000.

The results in Fig. 15 compare different wide-swath imaging methods. As shown in Fig. 15(a), CSA applied to the full echoes suffers from strong interswath interference, where overlapping signals clutter the reconstructed SAR image and obscure target features. Fig. 15(b) presents the ISTA reconstruction with full data, in which interswath interference produces prominent ghost images. The red boxes and arrows highlight the regions, where these ghost images are most concentrated. In contrast, the proposed method in Fig. 15(c) effectively suppresses such ghosting while preserving fine structural details, demonstrating its capability for wide-swath imaging over large scenes and its robustness to varying swath sizes.

D. Ablation Study

To validate the effectiveness of each module proposed in our joint optimization method JointSAR, we conducted several ablation experiments. Note that the experiments in this section were conducted using simulated data.

1) Impact of Nonuniform Pulse Patterns on SAR Imaging Design: In this section, we provide further insights into how the proposed method incorporates nonuniform pulse transmis-

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TABLE XI
PERFORMANCE COMPARISON OF DIFFERENT TRANSMISSION PATTERN CONFIGURATIONS TO VALIDATE THE IMPACT OF NONUNIFORM PULSE PATTERNS ON SAR IMAGING

Transmission Pattern Configuration	NMSE	PSNR	TBR	ENT
Matched transmission pattern	0.0067	58.25	63.72	5.59
Mismatched: Poisson disk	0.0112	56.09	55.50	5.67
Mismatched: Uniform down-sampling	0.0105	56.61	57.41	5.67
Random transmission pattern	0.0091	56.85	60.65	5.59

sion patterns into SAR imaging design and the benefits it brings to the system.

Existing SAR imaging methods can be categorized into traditional iterative methods and data-driven methods. Traditional iterative methods [38], [40] are inherently unable to learn the characteristics of nonuniform transmission patterns, as they apply identical processing steps to different pulse transmission patterns. In contrast, data-driven methods [12], [14] involve training an SAR imaging model, which inevitably becomes associated with a specific transmission pattern. As a result, the SAR imaging model learns the unique characteristics of the designed pulse transmission pattern.

To validate this hypothesis, several comparative experiments were conducted and shown in Table XI, and all the results are the averages obtained from ten independent trials to ensure statistical reliability. The first row in Table XI represents the performance of our proposed method, where an SAR imaging algorithm is trained with a matched pulse transmission pattern and evaluated using the same pulse transmission pattern. The second and third rows in Table XI show results when the trained SAR imaging algorithm is evaluated with mismatched pulse transmission patterns, specifically Poisson disk and uniform downsampling patterns, respectively. It can be observed from Table XI that the imaging quality significantly improves when the SAR imaging algorithm matches the transmission patterns, with PSNR increased by 2 dB and TBR increased by 8 dB. The last row shows that the pulse transmission pattern is randomly generated in each training iteration, effectively decoupling the SAR imaging algorithm from any specific transmission pattern. The results demonstrate that this training strategy of random transmission pattern fails to capture the characteristics of any specific transmission pattern, leading to suboptimal performance. This also highlights the advantages of training with a matched transmission pattern for the SAR imaging algorithm.

The above analysis demonstrates that data-driven methods enable the SAR imaging algorithm to learn the characteristics of specific pulse transmission patterns. In addition, nonuniform transmission patterns exhibit superior antialiasing properties compared with uniform patterns, and the learned transmission patterns outperform randomly generated nonuniform patterns, which will be further substantiated in Section IV-D.2.

2) Learnable Pulse Transmission Pattern: This section demonstrates that learnable nonuniform pulse transmission patterns achieve superior performance compared with fixed nonlearnable transmission patterns, whether nonuniform or uniform. Moreover, the sensitivity of the results to various pulse initialization strategies is also evaluated.

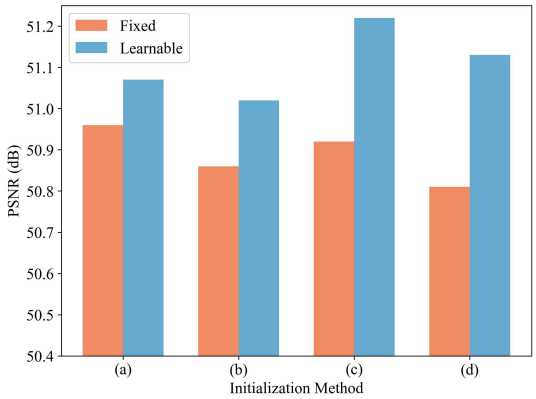

Fig. 16. Ablation experiments on whether to use the learnable pulse transmission pattern and robustness analysis of different initialization methods. Initialization method of the random part of the transmission intervals $\mathbf{d}_\eta$. (a) Uniform intervals, (b) random intervals following an exponential distribution, (c) random intervals following a normal distribution, and (d) random intervals following a uniform distribution.

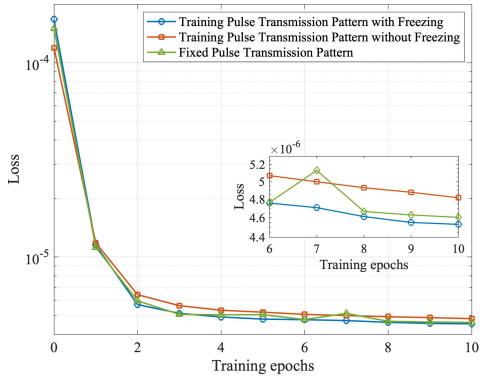

Fig. 17. Training loss curves comparing training the pulse transmission pattern with freezing, training it without freezing, and using a fixed pulse transmission pattern without training.

First, we employed four distinct initialization strategies for $\mathbf{d}_\eta$. These strategies mainly comprised uniform and nonuniform distributions, with the latter including exponential, normal, and uniformly distributed variations. For each initialization method, we implemented both fixed and learnable configurations. "Fixed configurations" indicates that the transmission pattern remains as the initialized sequence and is not optimized during training, while "learnable configurations" follow the joint optimization framework proposed in this article.

Table XII presents the experimental results of eight training configurations of $\mathbf{d}_\eta$, with a visualization of the PSNR metric provided in Fig. 16. From the experimental results, it can be observed that regardless of the initialization method, the learnable pulse transmission pattern consistently yielded superior image quality compared with the fixed pulse transmission pattern. Moreover, the choice of initialization method exhibited minimal influence on the final image quality, indicating the ability of our method to learn a good pulse transmission pattern and corresponding imaging parameters.

In addition, we analyze whether the dynamic changes in the learned pulse transmission sequence would affect the con-

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TABLE XII

ABLATION EXPERIMENTS ON WHETHER TO USE THE LEARNABLE PULSE TRANSMISSION PATTERN AND ROBUSTNESS ANALYSIS OF DIFFERENT INITIALIZATION METHOD OF THE RANDOM PART OF THE TRANSMISSION INTERVALS $\mathbf{d}_\eta$

	Loss		NMSE		PSNR		TBR	
Initialization method of $\mathbf{d}_\eta$	Fix	Learnable	Fix	Learnable	Fix	Learnable	Fix	Learnable
Uniformly intervals	5.05e-06	4.71e-06	0.0625	0.0597	50.96	51.07	49.02	53.49
Randomly intervals (exponential distribution)	4.84e-06	4.72e-06	0.0620	0.0601	50.86	51.02	51.41	52.85
Randomly intervals (normal distribution)	4.96e-06	4.57e-06	0.0624	0.0577	50.92	51.21	51.38	53.19
Randomly intervals (uniformly distributed)	4.80e-06	4.61e-06	0.0611	0.0586	50.94	51.13	46.15	52.60

TABLE XIII

ABLATION EXPERIMENTS ON THE METHODS OF TRAINING PULSE TRANSMISSION PATTERN $\mathbf{d}_\eta$

Learnable Variables (methods of learning)	Loss	NMSE	PSNR	TBR
Timings (standard learning)	4.80e-06	0.0602	51.00	51.60
Interval (standard learning)	4.71e-06	0.0596	51.07	54.58
Interval (random parameter flipping)	4.93e-06	0.0625	50.92	51.38
Interval (random parameter freezing)	4.57e-06	0.0577	51.22	53.19

vergence behavior of the SAR imaging algorithm. As shown in Fig. 17, comparing the blue curve (with learnable pulse transmission pattern) and the green curve (with fixed pulse transmission pattern), the convergence speeds are similar, indicating that the dynamic updates of the transmission pattern do not obviously hinder the optimization process. Moreover, the final training loss is lower when the pulse transmission pattern is learnable, demonstrating the effectiveness of jointly optimizing the transmission pattern and imaging algorithm.

3) Random Parameter Freezing: To validate the performance of random parameter freezing strategy before backpropagation, we designed four methods for ablation experiments, which can be broadly categorized into two main types: learning pulse transmission timings and learning pulse transmission intervals. Specifically, we use three approaches to learn the transmission interval: standard learning, random parameter flipping, and our random parameter freezing. The latter two methods introduce randomness into the pulse transmission intervals by either randomly flipping the gradients to their opposite sign or setting them to 0.

The results in Table XIII demonstrate that random parameter freezing yields better favorable outcomes across all evaluated metrics, including loss, NMSE, and PSNR. Direct learning of transmission intervals also proves effective, while random parameter flipping can lead to counterproductive results. Furthermore, learning intervals outperforms learning timings, consistent with the superior performance of the Poisson disk pattern over the Jittered pattern, as noted in [38].

Furthermore, we analyze the influence of the freezing probability factor p in (24) on the final imaging performance. The parameter p represents the proportion of retaining the gradients during each iteration, while the gradients of the remaining $1-p$ proportion of parameters are set to 0 to enhance robustness. When p is large, the strategy approaches standard optimization without freezing and introduces minimal perturbation, whereas a small p increases randomness significantly. At $p = 0$, the transmission parameters become completely random, which prevents meaningful gradient-based updates and leads to degraded performance. We varied p in the range [0, 1] and observed its impact on the imaging results. The

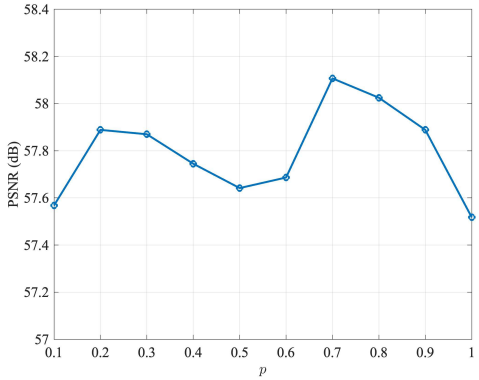

Fig. 18. Ablation experiments on random freezing probability factor p.

TABLE XIV

ABLATION EXPERIMENTS ON PULSE TRANSMISSION AND SAR IMAGING

Experiment Setting		Evaluation Metrics			
Signal transmission	SAR imaging	Loss	NMSE	PSNR	TBR
Fix	CSA	9.48e-05	1.2386	37.80	0.18
Learnable	CSA	7.40e-05	0.9756	38.86	3.16
Fix	Proposed method	4.96e-06	0.0624	50.92	51.38
Learnable	Proposed method	4.57e-06	0.0577	51.21	53.19

PSNR performance for different values of p is shown in Fig. 18. It can be observed that the performance tends to be lower when p is close to 0 or 1, while the best performance is achieved around $p = 0.7$. This setting preserves a portion of the gradients for effective optimization while introducing an appropriate level of randomness to enhance the robustness.

In addition, we plot the training loss curves over 10 epochs for both cases with and without the proposed freezing strategy to evaluate whether it affects the convergence of the training process. As shown in Fig. 17, comparing the blue curve (with freezing strategy) and the red curve (without freezing strategy), the overall convergence speeds are similar. However, the training loss with the freezing strategy is consistently lower, especially in the later epochs, indicating improved optimization performance without increasing the required training times.

4) Joint Optimization Mode: Finally, we performed an ablation experiment on the joint optimization of the pulse transmission pattern and imaging algorithm. By fixing parameters in either the pulse transmission intervals or the imaging algorithm to a nonlearnable mode, the joint optimization process was disrupted to evaluate its impact.

Table XIV presents the comparative results, showing that the joint optimization of both the learnable pulse transmission

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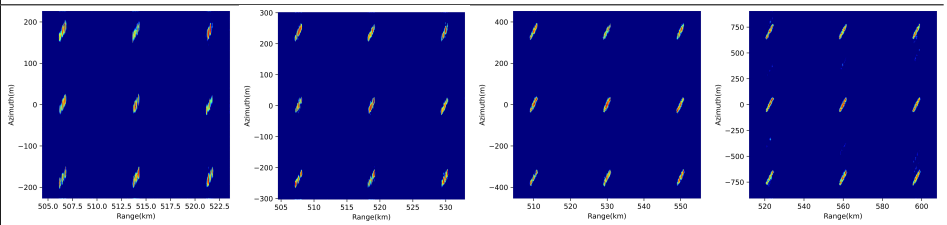
TABLE XV
UNIVERSALITY ANALYSIS OF SAR IMAGING RESULTS UNDER DIFFERENT MAXIMUM UNAMBIGUOUS WIDE SWATH SETTINGS [THE MODEL IS TRAINED WITH (B) AND TESTED WITH (A)–(D)]

	(a)	(b)	(c)	(d)
PRF	8000	6000	4000	2000
Wide swath before expansion (km)	9.75	16	28.5	66
Wide swath after expansion (km)	19.12	28.5	47.25	103.5
Azimuth resolution (m)	0.89	1.18	1.78	3.55
SAR imaging result after swath expansion				

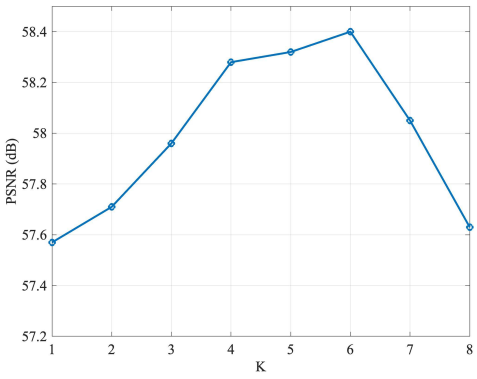


Fig. 19. Ablation experiments on different unrolling depths K.

pattern and learnable imaging algorithm yields better performance. While the improvement from the learnable sampling pattern alone is less pronounced, it still contributes to enhanced results when combined with learnable imaging algorithms. This underscores the effectiveness of joint optimization, where the most significant performance gains are achieved only when both components are optimized together.

5) Unrolling Depth K: To investigate the influence of the unrolling depth K on the SAR imaging performance, we conducted additional experiments by varying K from 1 to 8. As shown in Fig. 19, the PSNR increases rapidly when K increases from 1 to 4, while the improvement becomes marginal between $K = 4$ and $K = 6$. Notably, the performance slightly decreases when K increases further from 6 to 8. This degradation may be attributed to over-parameterization and optimization difficulties when the network depth becomes too large relative to the data scale. In contrast, too small K limits the representational capacity of the unrolled network. Considering that the number of model parameters increases linearly with K, we set $K = 4$ to achieve a good tradeoff between SAR imaging performance and computational complexity.

E. Universality Analysis

Since the proposed JointSAR method is typically trained with fixed parameters, assessing the universality of the proposed method to variations in radar parameters and scenario settings, such as maximum unambiguous swath widths, the slant range of the scene center, beam squint angle, and different levels of noise intensity, is crucial for real-world SAR applications. This section focuses on the algorithm universality analysis concerning radar parameters and varying noise intensities. All the data in this section are from the simulated test dataset.

To evaluate the universality of our proposed method, we test the trained model under various maximum unambiguous wide-swath settings without retraining. As shown in Table XV, the model is initially trained with PRF = 6000 Hz (b), and then tested with PRF of 2000, 4000, 6000, and 8000 Hz. These settings correspond to different maximum unambiguous swath widths and azimuth resolutions, as shown in Table XV. To adapt the trained model to different PRF settings corresponding to various maximum unambiguous swath widths, we directly scale the learned pulse transmission sequence as $\mathbf{T}_\eta \times (\mathrm{PRF}_{\mathrm{train}})/(\mathrm{PRF}_{\mathrm{test}})$. This is possible because our model does not rely on absolute timing, but rather learns a normalized pulse interval sequence $\mathbf{T}_{\mathrm{norm}} = (\mathbf{T}_\eta)/(T_{\mathrm{min}})$, which captures the relative structure of the transmission pattern. As a result, our model can generalize across different PRFs while preserving its essential ability to suppress aliasing artifacts. To ensure consistent visualization, the test scenarios are also proportionally scaled according to the PRF settings and the corresponding swath width configurations.

Using the adapted transmission sequence and our pretrained SAR imaging algorithm, we obtain the SAR imaging results, as shown in Table XV. Comparing Table XV—(a) and (c), when the swath width is moderately extended, the proposed method maintains good performance with effective suppression of aliasing artifacts and high image quality. In contrast, Table XV—(d) shows that when the swath width is further extended to approximately 100 km, noticeable azimuth aliasing appears. In this case, retraining our JointSAR is necessary to achieve higher imaging quality and effectively mitigate the aliasing artifacts. To further evaluate the universality of the proposed method under varying swath widths, Fig. 20 presents the PSNR curves comparing the SAR imaging result and the label. The results show that, although the model is trained

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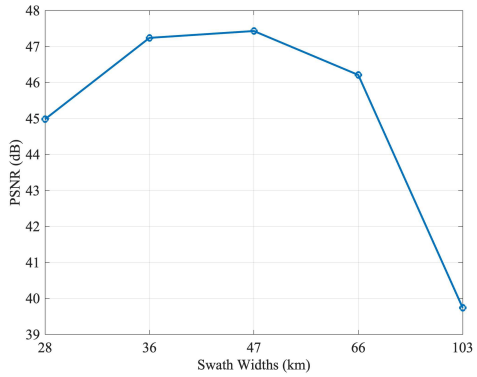

Fig. 20. Universality analysis with respect to PRFs and corresponding swath widths.

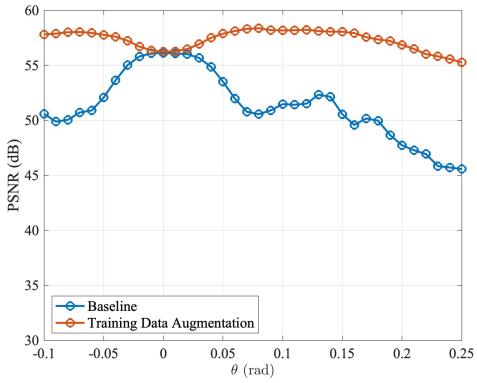

Fig. 22. Universality analysis with respect to squint angle θ. The blue curve represents the results trained directly with a specific θ, while the red curve shows the results after augmenting the training data.

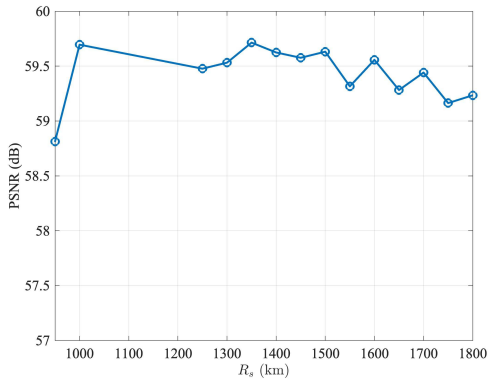

Fig. 21. Universality analysis with respect to the scene center range R_s.

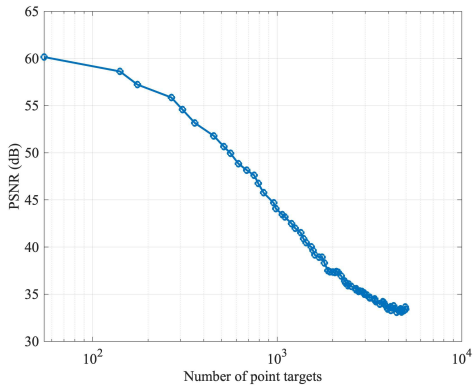

Fig. 23. Universality analysis with respect to different numbers of points.

at a swath width of approximately 28.5 km, it generalizes well to swath widths ranging from 28 to 66 km. Within this range, PSNR remains at a satisfactory level, indicating that the proposed method effectively mitigates aliasing artifacts caused by nonuniform sampling while maintaining high-quality wide-swath imaging without the need for retraining. However, when attempting to extend the swath width to around 100 km, the performance of aliasing suppression significantly deteriorates, resulting in a noticeable drop in PSNR. In such cases, retraining is required to ensure high-quality imaging and effective aliasing suppression.

Next, we evaluate the universality and robustness of the algorithm with respect to variations in the slant range distance of the scene center, R_s. During training, R_s was set to approximately 1×10^6 m. As shown in Fig. 21, the imaging quality remains stable across a wide range of R_s values, with only minor fluctuations in PSNR, indicating that the proposed method is robust to modest changes of the R_s parameter.

Then, imaging quality under different beam squint angles was compared. During training, the beam squint angle was fixed at -0.0277 rad, and the results for directly testing other beam squint angles are shown in the blue curve in Fig. 22. It can be observed that the performance remains relatively strong around the training angle, with a noticeable decline at other angles. Therefore, we perform data augmentation by incorporating different beam squint angles during training and also show the test results as the red curve in Fig. 22, which indicates a significant improvement in algorithm robustness after augmenting the training data.

In addition, we evaluate the limitations of the proposed method under varying scene complexity levels. Here, the scene complexity is approximately characterized by the total number of point targets in the scene, with the scene size fixed at 512×512. A larger number of targets means greater scene complexity. As shown in Fig. 23, while the proposed method achieves high PSNRs in sparse and moderately complex scenes, its performance gradually declines as the scene complexity increases due to stronger intertarget interference and the increased number of nonzero elements.

Finally, we analyzed the imaging quality under different simulated SNR levels. The training process included noise at 10 dB, while testing involved noise levels ranging from -5 to 30 dB. The results in Fig. 24 demonstrate that the algorithm maintains strong performance when the SNR is above 5 dB. At SNR levels below 5 dB, distinguishing between signal and noise becomes challenging, leading to a sharp decline in performance, which is consistent with our expectations.

F. Runtime Analysis

Table XVI presents the average runtimes of different SAR imaging methods, evaluated under the same computational

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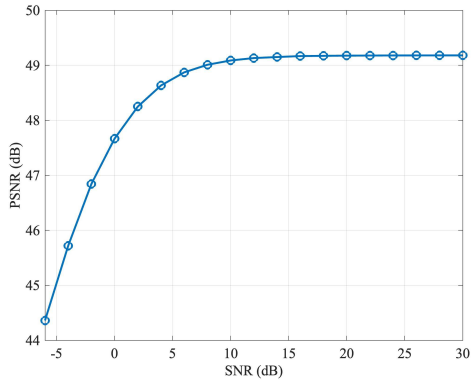

Fig. 24. Universality analysis with respect to different SNR levels.

TABLE XVI
COMPARISON OF INFERENCE TIME OF DIFFERENT METHODS

Method	Time(s)
Full CSA	0.0010
Uniform CSA	0.0007
Uniform ISTA	0.0656
Poisson disk ISTA [38]	0.0220
R-V Omega-K FISTA [40]	0.0216
DUN-AO-Net [14]	0.0139
Proposed	**0.0064**

environment to ensure a fair comparison. The results in Table XVI show that our proposed imaging method operates within the same order of magnitude in terms of runtime as the traditional CSA method and existing deep learning method [14]. The CSA method involves only a single forward propagation step, making it the fastest among all the methods. The runtime of uniform CSA methods is theoretically faster than full CSA due to the reduced data volume.

Overall, the computational complexity of the proposed SAR imaging algorithm is approximately the product of the number of unrolling depths and the combined cost of CSA forward operations (**R** module) and convolutions (**N** module), as detailed in Section III-D. According to the analysis in Section III-D, the computational complexity of convolution in **N** module is comparable to that of a single CSA forward operation. Therefore, the proposed DUN-based SAR imaging algorithm can achieve faster speeds compared with iterative algorithms due to reduced iterations.

V. CONCLUSION

In this article, we propose JointSAR, a novel joint optimization method that integrates a learnable pulse transmission pattern with a learnable imaging algorithm for high-resolution wide-swath SAR imaging. Specifically, the random parts of the existing nonuniform pulse transmission intervals are replaced by learnable variables, enabling us to overcome the limitations of existing random sequences and achieve superior SAR imaging performance. In the SAR imaging part, we propose a DUN-based imaging method with approximate observation operators to achieve high-resolution imaging. The final SAR imaging quality is used as the joint optimization objective to guide the update of all the learnable parameters. We also introduce a random parameter freezing strategy during back-propagation to improve the robustness of transmission timings. Extensive simulation and real data validation experiments and universality analyses demonstrate the effectiveness of the proposed method in compared with existing commonly used methods.

While the proposed method effectively learns improved pulse transmission patterns, its performance remains limited in complex imaging scenarios due to increased intertarget interference and the increased number of nonzero elements. Future work will focus on addressing these limitations by designing advanced algorithms for complex scenarios.

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